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Cue Classification in Preterm Infants Using Vision-Language Models

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Throughout the writing phase, I made considered and responsible use of large language models (LLMs) for refining the phrasing of my arguments and to enhance the clarity of my reasoning. All scientific content, arguments, and interpretations presented in this thesis remain my own and have been critically reviewed to ensure academic integrity.

Abstract

Premature infants in the Neonatal Intensive Care Unit (NICU) require continuous monitoring to ensure responsive and personalized care. Behavioral cues such as facial expressions and hand movements offer non-invasive insights into an infant’s physiological states but are currently assessed manually. This thesis explores the feasibility of using Vision-Language Models (VLMs) to automate the classification of these cues from video recordings. Motivated by the challenges of data scarcity in clinical settings, the study leverages the pretrained knowledge embedded in VLMs through a zero-shot, text-based classification pipeline based on predefined linguistic prompts. Using multi-label annotations adapted from a small dataset of five preterm infants recorded at UMC Utrecht, three experiments are conducted: (1) an embedding space analysis to assess semantic clustering, (2) a prompt-guided text classification method evaluated to determine label presence, and (3) a prompt ensembling strategy to enhance robustness. Results indicate that while model performance remains below clinical applicability thresholds, VLMs can discern visually salient cues, especially when guided by well-designed prompts. These findings demonstrate the potential of VLMs for behavioral cue recognition in data-scarce settings and suggest directions to further enhance performance through soft prompting, multilabel classification, and domain adaptation.

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1 Introduction

1.1 Problem Statement

Premature infants have a high need for special care due to their heightened vulnerability to health complications and developmental challenges. Because of this, they are commonly admitted to the Neonatal Intensive Care Unit (NICU). During their stay in the NICU, infants require continuous and personalized care, which is ideally optimized to match their exact needs. To achieve this, they are monitored for behavioral cues that provide insights into various physiological states, such as hunger, discomfort, sleep, and pain. Example cues expressed by infants include subtle facial expressions, limb movements, and changes in breathing patterns. Proper interpretation of these cues can help us understand an infant’s condition in a non-invasive way, and adapt care accordingly. This approach can ensure better scheduling of interventions, and accelerate progress in key areas, such as earlier hospital discharge [Kamran et al., 2020], and better detection of infant stress or discomfort [Holsti et al., 2005].

Various scoring systems have been developed to quantify these behavioral cues [Bik et al., 2022, Witt et al., 2016]. Ideally, infants would be scored continuously in order to ensure care is responsive to their specific needs. However, the effectiveness of existing scoring systems depends on accurate, real-time assessments by trained NICU nurses. In order to make cue-based care more attainable, the NICU could benefit from an automated assessment system. This would allow for dynamic adaptation of care practices without relying solely on periodic manual evaluations by NICU staff.

Many of the aforementioned physiological states can already be measured directly by techniques such as electrocardiography (ECG) [Zhang et al., 2023] and electroencephalography (EEG) [Lavanga et al., 2020], which rely on adhesive electrodes. However, these techniques can be costly, take time to set up, and — most notably — can damage the fragile skin of premature infants [Afsar, 2009]. Moreover, some states may not be able to be measured through the use of electrodes at all. For example, distinguishing active sleep from wakefulness using ECG data alone remains a challenging task [Zhang et al., 2023]. Alternatively, we can utilize unobtrusive camera recordings in order to classify infant cues, which can inform us about the physiological states of the infant accordingly. This addresses the problem of requiring continuous observation, and can provide complementary insights to existing methods that rely on biopotential signals.

Previous applications of computer vision in the NICU have demonstrated the ability for AI systems to improve clinical care. Research shows promising applications in non-contact vital sign monitoring devices by monitoring human tissue [Nagy et al., 2021], and in

some cases, accurately predicting apnea when the ECG monitor does not [Gibson et al., 2019]. Other applications include pain assessment [Zamzmi et al., 2019a] and detection of cerebral dysfunction [Gleason et al., 2024]. These systems build on more trivial computer vision applications, such as pose estimation and face detection, which rely on vast amounts of data. However, in neonatal care, video data is often scarce. Due to privacy concerns, there exists no public dataset, making hospitals reliant on their own data. Moreover, data quality can be unreliable due to the volatile nature of the environment. Differences between infants in lighting, skin color, behavior, and care, introduce a high variance between collected videos. The scarcity of video data in addition to a high data variance makes it challenging for machine learning techniques to generalize effectively over the available data [Staeb, 2023].

Vision-text alignment models, such as CLIP [Radford et al., 2021], align visual and textual representations in a shared embedding space. Due to the vast amount of language data used during the pre-training stage of these models, they show very promising capabilities in generalizing across diverse visual domains. By utilizing such a pre-trained model for classification tasks, we may overcome the data scarcity and variance issues mentioned earlier. Moreover, Vision-Language Models (VLMs) commonly incorporate CLIP or similar models [Liu et al., 2024], and are frequently designed for visual question answering (VQA) tasks. These models can be utilized to introduce domain-specific prompts to the embedding space. Several studies have demonstrated that enriching the textual input, either by augmenting their prompts [Mirza et al., 2024] or their class labels [Saha et al., 2024], can significantly enhance zero-shot classification performance in VLMs. These improvements have been observed both when using CLIP directly and when employing a linear classifier built on top of VLM-derived features [Abdelhamed et al., 2024].

Following the outcomes of this research, we suggest that by integrating linguistic features into image classification through domain-specific prompts, we can create a richer vision-text embedding that enhances the model’s understanding of the task. Unlike conventional classification approaches, which operate purely on visual data, this enriched embedding leverages language to provide contextual depth, potentially leading to improved recognition and interpretation of subtle infant cues. By enabling models to reason over visual input with the guidance of natural language, this approach could enable more robust and automatic assessment of neonatal behavior and physiological states, ultimately supporting better decision-making in the NICU.

1.2 Contributions

The primary goal of this research is to explore the feasibility of using Vision-Language Models (VLMs) for classifying behavioral cues in video data of preterm infants. We

hypothesize that introducing language alongside vision can yield richer, more generalized embeddings, ultimately improving the understanding and classification of intricate behavioral cues expressed by preterm infants. This goal is motivated by the overarching aim to contribute to better infant care and well-being in the NICU. The study will be guided by the following main research question:

How effective are Vision-Language Models at classifying behavioral cues in video recordings of preterm infants?

To address this question, a pre-trained VLM will be implemented and evaluated. The model input will consist of images extracted from annotated videos of preterm infants that are admitted to the NICU, as well as predefined textual prompts to guide the model’s output. The model’s performance will be assessed using labeled validation data. We will compare the model accuracy against baseline models, including a majority vote classifier and a prior probability-based classifier. We argue that VLMs will be applicable for behavioral cue classification if their performance significantly surpasses these baselines. Additionally, precision, recall and F_1 will be measured to assess the model’s ability to correctly identify each behavioral cue. These metrics are crucial in determining the clinical utility of the model, as they provide insight into both sensitivity and specificity in recognizing preterm infant behaviors.

This research builds on the premise that pre-trained VLMs will perform reasonably well for classification tasks without additional training, due to their large, pretrained latent space. As such, our approach does not involve fine-tuning. Our approach leverages the model’s text-output capabilities by comparing its predictions to a set of textual answer templates using a text similarity metric. This enables us to confine the model’s output to a distinct set of predefined classes and compute a vector of probabilities representing the correspondence of the model’s output to each class. Consequently, this can make the model sensitive to the composition of the input prompt as well as the answer template. This motivates the following sub-question:

What impact do different compositional and linguistic design choices of input prompts and answer templates have on the classification performance of the Vision-Language Model?

In order to answer this question, we will explore the difference in model performance between different prompt and template designs. Specifically, we will explore the impact of the following factors:

- *Providing task context vs. no context:*

Based on findings by [Radford et al. \[2021\]](#), providing more context to the class labels may improve alignment between the model’s text output and the image, leading

to more accurate classification. We hypothesize that a similar effect may occur in our text-based classification setting, where providing task-relevant context in the prompt could enhance the semantic match between the model’s output and the predefined answer templates, thereby improving classification accuracy.

- *Open-ended vs. closed questions:*

Open-ended questions may result in more evenly distributed prediction probabilities, whereas closed questions could constrain responses, potentially increasing model confidence. To establish a meaningful baseline, we also include a no-prompt condition to assess the model’s default performance in the absence of linguistic guidance.

By investigating both the overall performance of VLMs in classifying behavioral cues and the influence of linguistic prompt design, this research aims to provide foundational insights into the applicability of vision-language models in neonatal care. The findings may contribute to the development of more interpretable, accessible, and non-invasive monitoring solutions, ultimately supporting more responsive and specialized care in the NICU.

2 Related work

This section provides an overview of the relevant medical, technological, and methodological foundations underpinning this research. We begin by outlining the developmental risks associated with preterm birth and the role of behavioral cues in monitoring infant well-being. We review existing approaches for behavioral assessment in neonatal care and highlight the limitations of manual cue classification. We continue by examining how computer vision has been applied in the NICU in relation to infant monitoring, and introduce VLMs as a promising alternative for automated cue interpretation. Finally, prompting strategies and alternative classification techniques using VLMs are discussed to support the methodology proposed in this thesis.

2.1 Preterm Infants

Preterm infants, born before 37 weeks of gestational age (GA), face an increased risk of various health complications due to their early arrival. Preterm birth is the leading cause of neonatal mortality worldwide, with late preterm infants experiencing mortality rates three to five times higher than those of term infants (≥ 37 weeks) [Bulut et al., 2016]. Moreover, the risk of mortality rises as GA decreases [Khasawneh and Khriesat, 2020]. Despite advancements in neonatal care, which have significantly improved the survival rates of extremely preterm infants [Bell et al., 2022], preterm birth remains associated with long-term complications and higher morbidity rates than term infants [Allen, 2008]. An estimated 10% of preterm infants develop neurodevelopmental impairments [Blencowe et al., 2013], including language, cognitive, sensory, and motor deficits [Platt, 2014].

The most commonly acknowledged cause of these impairments is biological, as preterm birth disrupts a crucial period of brain development and maturation of neural structures [Song, 2022]. However, it is frequently theorized that postnatal environmental factors also play a significant role in neurodevelopmental outcomes. For example: sleep, hunger, and pain are key aspects influenced by neonatal care. Research shows that infants with higher proportions of active sleep exhibit greater white matter development, a critical marker of neural connectivity [Wang et al., 2024c]. This suggests that sleep disruption due to frequent medical interventions and noise in the NICU can impair neural connectivity. Additionally, pain and stress in the NICU are also believed to be contributors to developmental deficits in preterm infants. Studies have shown that neonatal stress is linked to adverse developmental outcomes at a corrected age of 24–30 months [Pavlyshyn et al., 2024]. Lastly, nutrition plays a crucial role in brain development. Preterm infants who experience malnutrition are at risk of cognitive impairments, including lower intelligence quotients (IQ) [Keunen et al., 2015]. Neuroimaging studies have demonstrated that early malnutrition can lead to

persistent alterations in brain volume and white matter integrity [Coviello et al., 2018], further emphasizing the long-term impact of inadequate nutrition.

2.1.1 Behavioral Cues

The findings outlined in the preceding section highlight some of the environmental factors that can play a critical role in the development of preterm infants. In order to mitigate the risks associated with preterm birth, early detection of potential complications is crucial. A non-intrusive method for assessing the well-being of preterm infants can be performed by interpreting their behavioral cues. Several studies have developed and validated scoring systems to analyze infant states and needs based on behavior. These scoring systems vary widely, assessing parameters such as eye movements, body movements, facial expressions, sounds, and respiratory patterns.

One of the most extensive behavioral assessment tools is the Assessment of Preterm Infant Behavior (APIB) [Als et al., 1982], which is derived and adapted from the Neonatal Behavioral Assessment Scale (NBAS) [Brazelton, 1973]. The APIB is designed to examine the functionality of various subsystems in the infant and evaluates five key domains: physiology, motor activity, state organization, attention/interaction, and regulatory capacity. These assessments occur before, during, and after specific maneuvers are administered to the infant. Assessments include observing the respiratory system, posture, sleep-wake states, alert-state, and avoidance or groping behavior (e.g. finger splaying, yawning, tongue extension hand-to-mouth movement). The APIB is applicable once a preterm infant reaches a developmental stage where handling without extensive medical equipment is feasible. However, it is not validated whether the states assessed using APIB directly correlate with those observed in the NICU.

For earlier and more specialized assessment, several validated scoring systems have been developed. Scoring the sleep states of preterm infants has been thoroughly studied by Prechtl [1974], whose suggested system shares similarities to the APIB. Another study introduces BeSSPI, which refines aspects of Prechtl’s method and other sleep scoring approaches, providing an easy-to-learn, validated scoring framework [De Groot et al., 2022]. The BeSSPI incorporates temporal behaviors such as rhythmic or reflexive facial and body movements, along with sound production and heart rate variability. Beyond sleep regulation, behavioral cues can indicate an infant’s need for nourishment. Common hunger cues include crying, sucking and rooting [Puckett et al., 2008]. Lastly, several scoring systems have been developed to classify pain. The Neonatal Facial Coding System (NFCS) evaluates pain based on specific facial expressions, including brow bulging, eye squeezing, naso-labial furrowing, and mouth opening [Grunau et al., 1998]. Alternatively, the Neonatal Infant Pain Scale (NIPS) assesses pain through a combination of facial

expressions, limb movements, and state changes, including cry intensity and breathing patterns [Hudson-Barr et al., 2002].

Behavioral cue classification is often deemed inefficient, as it requires both extensive expert training and time taken for observation. This suggests the need for an automated approach. However, several inherent limitations are at play when developing such systems. Firstly, behavioral expressions are expressed differently with respect to the infant’s age [Als et al., 1988], leading to variations in how cues manifest between infants. Additionally, some cues described earlier are temporal and unfold over a period, while others are more static, requiring different classification systems. The intensity of responses also varies significantly. Sometimes behaviors may be pronounced, whereas at other occasions may be extremely subtle, making them difficult for vision-based systems to detect and classify accurately. Another critical challenge is the ambiguity of certain infant cues. Behaviors such as mouthing, mouth opening, and tongue protrusion can be difficult to distinguish, leading to potential misclassification by both human observers and automated models. Furthermore, infants may express multiple cues simultaneously, increasing the complexity of accurate classification and interpretation. These limitations underscore the difficulty of developing a robust and reliable means of classification, whether manual or automated.

2.2 Computer Vision in the NICU

The application of computer vision (CV) in neonatal care has been extensively explored [Keles and Bagci, 2023, Krbec et al., 2024, Rallis et al., 2024], offering a non-invasive and real-time monitoring solution. CV holds the potential to reduce the need for intrusive procedures while improving the accuracy and timeliness of clinical decision-making.

Many studies have investigated the use of CV for monitoring physiological parameters in neonates. For instance, heart rate can be inferred by detecting subtle color changes in the skin caused by blood volume fluctuations with each heartbeat [Gibson et al., 2019, Wu et al., 2012]. However, this method alone is not always reliable due to its susceptibility to variations in lighting conditions. Similarly, respiratory rate can be estimated using CV techniques, such as optical flow analysis of chest movements combined with machine learning. One study successfully employed optical flow measurements from chest movements alongside machine learning algorithms to estimate respiratory rate [Nagy et al., 2021]. Nevertheless, challenges such as occlusions from blankets and medical equipment can impact accuracy. Some studies have attempted to address these limitations by incorporating radar imaging to measure respiratory rate [Kim et al., 2019].

In addition to physiological monitoring, CV has been explored for facial analysis in neonates. Several studies have investigated face detection [Dosso et al., 2022, Hausmann et al.,

2022] and facial landmark detection [Grooby et al., 2023]. Researchers have leveraged pre-trained convolutional neural networks (CNNs) and YOLO-based architectures to classify neonatal facial expressions for assessing pain [Zamzmi et al., 2019b], jaundice [Mansor et al., 2012], and sleep-wake states [Awais et al., 2021]. However, face detection and facial landmark identification in neonates remain challenging, as existing architectures are primarily trained on adult faces, which do not generalize well to neonatal facial characteristics [Wan et al., 2022].

Another area of interest in neonatal CV research is behavioral monitoring. Pose estimation techniques have been applied to classify sedation levels and detect signs of cerebral dysfunction [Gleason et al., 2024]. However, accurate pose estimation in neonates is hindered by occlusions from medical equipment and blankets, suboptimal camera positioning, and the scarcity of comprehensive datasets tailored to infant body proportions. Some studies suggest that newer architectures, such as vision transformers (ViTs) and 3D convolutional neural networks (3D-CNNs), could enhance pose estimation performance. However, these models require substantially large datasets to achieve robust generalization [Staebe, 2023]. Additionally, the collection and storage of neonatal video data raise privacy concerns, prompting researchers to explore alternative approaches. For instance, depth imaging has been proposed as a privacy-preserving alternative to conventional RGB cameras while still capturing infant poses [Moccia et al., 2019].

2.3 Vision-Language Models

A Vision-Language Model (VLM) is a machine learning model that is designed to process both visual and textual data. VLMs are built by combining a Large Language Model (LLM) with a vision encoder, essentially introducing an extra modality to the LLM. VLMs are therefore often referred to in literature as multimodal LLMs [Wang et al., 2024a]. Which is the category of models that process multiple modalities of data (e.g. language, vision, sound, and more) and output natural language. A VLM processes an image and a corresponding query about the content of the image, written in natural language. The model then generates a textual response that answers the question by reasoning about the relationship between the image and the given prompt. The LLMs used in VLMs are commonly pre-trained on extremely large datasets. This makes them very flexible and applicable to various use cases by simply modifying the textual prompt. This architecture has been successful for tasks such as image captioning [Li et al., 2023] and visual question answering (VQA) [Liu et al., 2024, Alayrac et al., 2022].

VLMs designed for VQA typically rely on a strong foundational vision-text alignment model such as CLIP [Radford et al., 2021] or ALIGN [Jia et al., 2021]. These models align images and text by encoding both modalities into embeddings using separate vision

and text encoders. These embeddings are mapped to a shared embedding space, enabling the model to understand and reason about the relationship between visual and textual inputs. This is achieved by optimizing the contrastive loss, ensuring that image-text pairs are close to each other in the embedding space while non-matching pairs are pushed apart. This shared space serves as a foundation for many downstream tasks, including image retrieval, zero-shot classification, and as a backbone for VLMs, where CLIP’s vision encoder is frequently adopted.

Besides the vision encoder, VLMs incorporate a textual encoder that converts text input into textual embeddings. Once both the visual and textual data are encoded into embeddings, they are projected into a shared embedding space that is interpretable by the language component. There are two main approaches on how this projection is done, Dai et al. [2024] refer to the approaches as decoder-only VLMs and cross-attention-based VLMs. The approach is named after where and how the visual and textual embeddings are combined into a shared embedding space.

2.3.1 Decoder-Only Models

In the decoder-only approach, vision embeddings are projected into the same dimensions as the text embeddings at an early stage. This transformation is performed by a projection layer, which is typically a simple linear layer. After being projected into the same dimension, the visual and textual embeddings are concatenated into a shared representation, which is then processed by a decoder-only LLM. A decoder-only LLM is a type of transformer model that consists solely of the decoder component of the original Transformer architecture [Vaswani, 2017]. The LLM transforms the shared embedding into a textual output. This method is particularly attractive due to its simplicity, as it requires minimal architectural modifications to existing decoder-only LLMs.

A notable implementation of this approach is LLaVA [Liu et al., 2024], which leverages vision embeddings extracted from the CLIP [Radford et al., 2021] vision encoder. These embeddings are projected into the text embedding space using a trainable linear layer before being concatenated to the textual embeddings. The combined embeddings are then passed onto a fine-tuned LLaMA decoder-only model [Touvron et al., 2023], which generates responses in natural language. Other models following this approach include Molmo [Deitke et al., 2024] and Qwen2-VL [Wang et al., 2024b]. Molmo is a nearly fully open-source VLM trained on its proprietary open-source dataset, PixMo. It uses a CLIP inspired vision encoder and a decoder-only architecture based on Vaswani [2017] as its LLM. Qwen2-VL, in contrast to LLaVA and Molmo, employs a modified ViT [Dosovitskiy, 2020] as its vision encoder rather than a contrastive vision-text alignment model such as CLIP or ALIGN. This ViT-based architecture enables Qwen2-VL to process images and

videos at their native resolutions while integrating the projection step within the model. During training, decoder-only models often pre-train the projection layer while freezing the LLM and the vision encoder. Qwen2-VL follows this paradigm, but because projection is integrated in Qwen2-VL’s ViT, it pre-trains the vision encoder instead. Molmo, however, omits the projection pre-training phase, instead relying on an initially higher learning rate to facilitate quick convergence. Qwen2-VL performs a similar intermediary step, unfreezing all parameters and utilizing higher diversity data in order to introduce more comprehensive learning. Fine-tuning for such models often involves keeping the vision encoder frozen while training the weights in the projection layer and LLM [Liu et al., 2024, Wang et al., 2024b]. During fine-tuning, many models, including LLaVA and Qwen2-VL, freeze the vision encoder while training the projection layer and language model [Liu et al., 2024, Wang et al., 2024b]. This is done to preserve the pre-trained visual representations of the vision encoder.

2.3.2 Cross-Attention-Based Models

Cross-attention-based VLMs employ an architecture that more closely aligns with the original Transformer framework proposed by [Vaswani, 2017]. In this approach, the vision encoder substitutes the encoder component of the transformer. The generated vision embeddings are subsequently integrated into the decoder through a cross-attention mechanism. Enabling the model to selectively focus on relevant parts of the visual input while generating text. Although implementing cross-attention-based VLMs is more sophisticated than implementing decoder-only models, they are generally more efficient than decoder-only models during both training and inference. Unlike decoder-only models, images do not have to be converted into lengthy token sequences before being passed to the LLM. Additionally, the cross-attention mechanism allows for more controlled information flow. This allows the model to query only relevant features and ensures that text-only performance is retained when there is no image input.

A noteworthy example of a cross-attention-based VLM is Flamingo [Alayrac et al., 2022], which is particularly notable for its strong few-shot learning capabilities. Flamingo encodes images or videos using a pre-trained CNN-type [Wu, 2017] architecture called Normalizer-Free ResNET (NFNET) [Brock et al., 2021]. The outputs of the vision encoder, which vary in size, are then passed through a perceiver resampler, which transforms them into a fixed-size embedding. Analogous to decoder-only models, the perceiver resampler essentially acts as a projection layer preceding the cross-attention module. From here, the processed vision embeddings are integrated with the language representations through a gated cross-attention block before being fed to the language component. A key advantage of Flamingo’s architecture is its ability to iteratively interleave images or videos with text.

This design directly contributes to its few-shot learning performance, as the model can efficiently process and integrate new visual information with minimal labeled training data.

In addition to Flamingo, other cross-attention-based VLMs include BLIP-2 and the multimodal LLaMA3 models [Li et al., 2023, Dubey et al., 2024]. Both models adopt a ViT as their vision encoder but differ in their approach to bridging visual and textual representations. Unlike Flamingo, which utilizes a CNN-based vision encoder followed by a perceiver resampler, BLIP-2 introduces a Querifying Transformer (Q-former) to bridge the vision and text encoders. The LLaMA-V models instead use a vision adapter that consists of a series of cross-attention layers. Although the multimodal LLaMA3 models implement a ViT as their encoder, their image-text pairs used for training are filtered based on their CLIP score, suggesting that CLIP’s success resides in its data rather than its architecture [Xu et al., 2023]. In both Flamingo and BLIP, the image encoder and language model remain frozen during pre-training. In Flamingo, only the perceiver resampler, cross-attention modules and feedforward layers are trained. Similarly, only the Q-former and a feedforward layer are trained for BLIP-2. Conversely, LLaMA3 models unfreeze the vision encoder during pre-training, while keeping both the cross-attention modules and language model frozen. During the fine-tuning phase, LLaMA3 selectively updates the vision encoder and image projection layers.

2.4 Prompting

Prompting serves as the primary interface here between users and VLMs, allowing users to provide context and instructions that direct the model’s output. Well-structured prompts can significantly improve classification accuracy, whereas poorly designed ones may lead to misinterpretations and decreased performance. Understanding different prompting approaches is therefore crucial to optimize VLM performance.

Prompting techniques can be categorized into hard prompts and soft prompts. Hard prompts consist of human-readable text, whereas soft prompts are continuous vectors that can be trained and which are not directly interpretable by humans.

2.4.1 Hard Prompts

Hard prompts are typically natural language instructions or examples provided to the model. Such prompts are easily understood by humans and do not alter the model’s parameters. The effectiveness of a hard prompt depends on its wording and structure, which often requires trial and error to achieve optimal performance. The CLIP study initially highlighted the importance of text composition for classification [Radford et al.,

2021]. The authors found that formulating class labels within natural language prompts significantly improves image classification accuracy. Instead of using a single-word label, structuring the prompt in a sentence format — such as “*A photo of a {label}.*” — yields better performance. Further improvements were observed when adding domain-specific information to the prompt, for example: “*A photo of a {label}, a type of pet.*” for classifying images from the Oxford-IIIT Pets dataset. This finding underscores the importance of contextual information in multimodal classification and has inspired further research into adaptive prompting strategies.

While a single prompt like “*A photo of a {label}.*” is a good general default, any individual phrasing might not capture all aspects of a class. Different wording or context can shift the text embedding slightly. Instead of relying on one prompt, CLIP therefore introduces prompt ensembling. With this approach, multiple prompt variants for each class are created and combined in embedding space. Each prompt acts as a slightly different interpretation of the class’s semantics. By ensembling them, the model can form a more robust textual representation that covers various contexts or meanings of the class. This reduces ambiguity and improves alignment with the image space.

Research by Mirza et al. [2024] takes this a step further by automating the entire process. Rather than relying on manually crafted prompts, they use a two-stage meta-prompting strategy to generate category-specific prompts. First, it instructs an LLM to generate task-specific but label-independent templates. Then, these templates are filled with class names to produce descriptive prompts. Finally, the resulting text embeddings are ensembled to form a single, robust class representation. Their strategy improves classification performance of CLIP while removing human prompt ideation from its pipeline.

2.4.2 Soft (Learnable) Prompts

Soft (or continuous) prompts are vectors inserted into the model’s input embeddings. Unlike hard prompts, they are constructed through training rather than written in natural language, which makes them uninterpretable for humans. During prompt tuning, the model’s original weights are frozen while only a small set of additional parameters representing the prompt are updated. Soft prompts can achieve similar better accuracy than the best hand-crafted text prompts [Li and Liang, 2021]. Lester et al. [2021] showed soft prompt tuning can match or surpass fine-tuning on language tasks with orders-of-magnitude fewer trainable parameters

In the vision-language domain, Context Optimization (CoOp) [Zhou et al., 2022c] extends this idea by learning continuous prompt vectors for CLIP’s text encoder. CoOp yields higher classification accuracy than manual prompts across many image recognition tasks. Notably, it achieves strong performance with as few as 1–2 labeled samples per class.

One weakness observed was that a prompt tuned on a specific set of classes may not generalize to unseen classes. Conditional Context Optimization (CoCoOp) addresses these limitations of CoOp [Zhou et al., 2022b]. It makes the prompt conditional to each image input instance, improving the generalization to new classes. Subsequent research also explores coupling visual and textual prompts. MaPLe [Khattak et al., 2023] jointly learns a visual prompt and a text prompt, optimizing them together for VLM fine-tuning. This achieved better results than tuning text prompts alone and, along with CoCoOp, suggests that prompts need not only be textual but can also depict regions of interest in images.

Researchers have also explored methods to dynamically generate adaptive prompts using LLMs. LLaMP [Zheng et al., 2024] is one such approach that integrates an LLM into the CLIP framework to generate more informative descriptors for images. LLaMP passes class labels through an LLM using a query prompt, producing richer descriptions that better capture class-specific nuances. This approach has shown particular effectiveness in few-shot classification tasks, where limited training examples make it challenging for CLIP alone to differentiate between visually similar classes.

2.5 Alternative Classification Techniques

By leveraging both visual and textual representations, VLMs enable more contextually informed classification decisions compared to purely vision-based architectures such as traditional CNNs and ViTs. The integration of semantic information through language allows VLMs to reason beyond visual similarity.

This idea of contextually informed classification is demonstrated by Saha et al. [2024], who enrich class labels with detailed textual descriptions generated by an LLM. These descriptions allow the model to associate images with a richer semantic space. This strategy leads to significant improvements in zero-shot classification performance, particularly in fine-grained domains. The effectiveness of this multimodal approach is further supported by the work of Abdelhamed et al. [2024]. They employ a VLM that generates additional descriptive feature vectors from a fixed set of hard prompts. The extracted features are combined with the original image in a linear classification model, resulting in improved performance compared to using the image alone.

Another innovative approach to classification with VLMs, CascadeVLM, involves a hybrid architecture that combines CLIP with a VLM [Wei, 2024]. CascadeVLM employs a two-stage classification process. The first stage employs CLIP to classify the top-k most probable candidate classes. In the second stage, the VLM is able to make a more fine-grained prediction between the candidate classes. This cascading approach effectively transforms a zero-shot classification problem into a few-shot classification task, where the

VLM only needs to differentiate among a small subset of possible classes rather than the entire dataset.

Beyond static image classification, VLMs have also been successfully adapted for action recognition tasks by encoding text prompts into the temporal dimension. Recent work by [Lu et al. \[2024\]](#) introduced a method where the visual encoding of an image is aligned with descriptive text outputs from a VLM. By integrating text-based representations — encompassing action categories, components, descriptions, and contextual information — into the visual embedding space, this method leverages the generalization capabilities of VLMs while preserving the spatiotemporal understanding of video data.

3 General Methodology

This section outlines the general methodology that precedes the experimental design. First, we describe the SLAPI dataset, including preprocessing steps, multi-label reformulation, and motivate our fold selection. Second, we motivate our choice of **Qwen2-VL-3B-Instruct** as the VLM used in our experiments.

3.1 Data

The dataset used to carry out this research originates from the SLAPI study, conducted at the University Medical Center (UMC) Utrecht. It comprises video recordings of five preterm infants admitted to the NICU. Participant recruitment and data collection were carried out by UMC Utrecht in accordance with ethical guidelines. The dataset was initially annotated for the study by [Staeb \[2023\]](#). Infants were included based on specific criteria: GA of at least 32 weeks, visible facial features, minimal occlusions, and the presence of observable behavioral cues. The five selected infants, consisting of both sexes, have a mean GA of 35.8 ± 1.6 weeks, ensuring a relatively homogeneous sample in terms of developmental stage.

Recordings are obtained using a consistent top-down camera angle, providing an unobstructed view of the infants’ upper bodies and facial expressions. This area is most relevant for behavioral cue detection. The duration of annotated video footage varies across participants, ranging from approximately 30 to 83 minutes per infant. Annotations were performed manually by research interns at UMC Utrecht. Prior to analysis, all videos were cropped to exclude the infants’ lower bodies, which are typically covered by blankets and do not contribute meaningful visual information for the classification tasks. The resulting cropped frames retained sufficient resolution and visual quality for downstream analysis. A detailed overview of the recordings per infant is provided in Table 3.1.

Infant	Gest. Age	Sex	Total Cues	Included Cues	Video Resolution	Video Duration	Annotation Span
1	37.4	M	18	11	884×792	1:36:47	1:01:00
2	35.7	F	19	10	994×1102	0:58:52	0:30:39
3	35.1	F	19	13	956×844	0:30:38	0:30:28
4	35.3	M	9	3	974×926	1:00:55	0:57:01
5	35.4	M	21	14	1180×906	1:27:40	1:23:16

Table 3.1: Recording characteristics of the included participants from the SLAPI study.

3.1.1 Preprocessing

Each video is manually annotated to identify instances of predefined behavioral cues exhibited by the infants. For every annotated segment, a representative frame was selected manually, capturing the moment where the behavior was most clearly expressed. This

approach helps ensure that the labels are assigned to frames that best capture the relevant behavioral expressions, minimizing ambiguity in the data.

The dataset remains limited, as data collection is still ongoing at the UMC Utrecht. It is therefore important that we consider only labels that generalize well over the annotated videos. To maintain label consistency and statistical robustness, several exclusion criteria were applied:

1. Labels that appear fewer than ten times across the entire dataset are excluded to prevent data sparsity.
2. Labels that occur in fewer than three of the five infant videos are removed to ensure generalizability across subjects.
3. Labels describing purely temporal information, such as head movement, hand movement, and arm movement, are not included, as they do not correspond to static frames suitable for model evaluation.

Following the application of these criteria, each annotated frame was reviewed manually by the author to validate label alignment with visual content. Misaligned or ambiguous labels were corrected or reannotated accordingly. The label set derived from these criteria is summarized in Table 3.2.

Classes	Infant 1	Infant 2	Infant 3	Infant 4	Infant 5	Annotations per Class
Flexing	4	0	1	4	1	10
Finger Splay	2	7	2	3	2	16
Eye Squeeze	11	0	3	3	7	24
Hand-To-Face	5	24	23	0	2	54
Hand-To-Mouth	4	15	16	1	11	47
Mouth Open	21	0	4	0	23	48
Adult Hand	25	20	2	0	14	61
Yawn	17	2	1	0	2	22
Bottle Feeding	6	1	0	0	18	25
Grasping	3	5	4	0	0	12
Hand Sucking	0	9	1	0	12	22
Tongue Out	19	3	2	0	6	30
Annotations per infant	117	86	59	11	98	Total Frames: 371

Table 3.2: Overview of the included number of annotations per class per infant from the SLAPI dataset.

3.1.2 Multi-Label Annotations

Although each frame in the original dataset was annotated with a single primary label, in practice, multiple behavioral cues often co-occur. To reduce ambiguity and better reflect the complexity of infant behavior, the dataset is reformulated as a multi-label classification problem. This reformulation enhances both the richness of the dataset and the robustness of the evaluation, by allowing multiple concurrent labels per frame. A

custom annotation script was employed to facilitate this process: it presented each frame for secondary inspection, allowing the annotator to identify and tag additional present cues. This resulted in a substantial increase in the total number of labels, as shown in the updated distribution in Table 3.3.

Classes	Infant 1	Infant 2	Infant 3	Infant 4	Infant 5	Annotations per Class
Flexing	5	1	2	4	3	15
Finger splay	6	18	10	4	4	42
Eye squeeze	14	15	5	3	18	55
Hand-to-face	12	64	48	7	33	164
Hand-to-mouth	5	47	33	6	25	116
Mouth open	59	42	11	0	44	156
Adult hand	32	22	2	0	32	88
Yawn	19	2	3	0	3	27
Bottle feeding	10	7	0	0	18	35
Grasping	20	8	5	1	1	35
Hand sucking	0	21	6	0	14	41
Tongue out	24	6	4	0	7	41
Annotations per infant	206	253	129	25	202	Sum: 815

Table 3.3: Overview of the included number of annotations per class and per infant after reformulating the SLAPI dataset into a multi-label format.

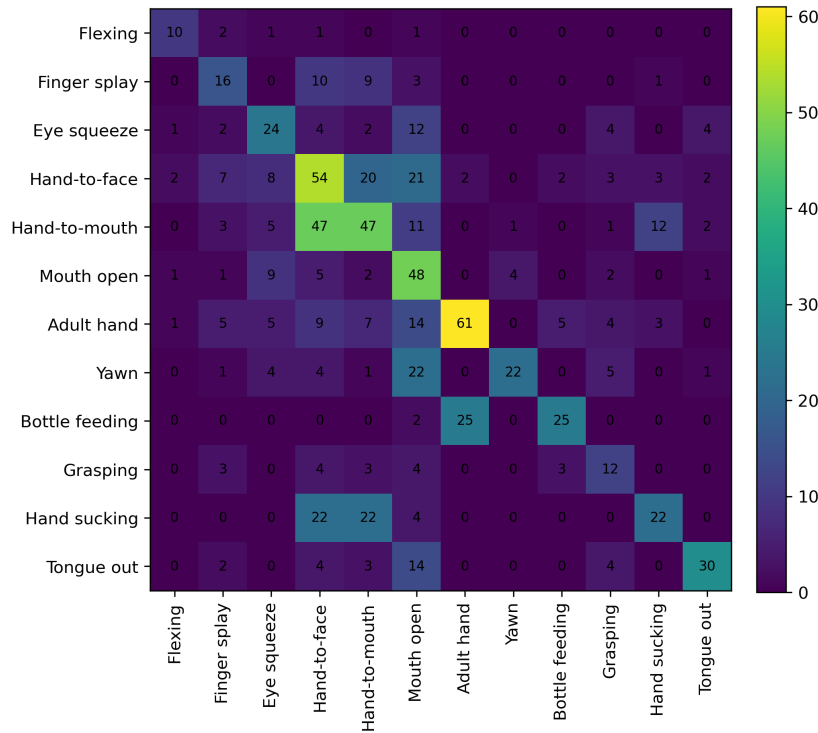


Figure 3.1: Transformation matrix of the original single-label annotated SLAPI dataset into a multi-label annotated dataset. Each cell denotes the number of times a cue on the X-axis was added to the original sample annotated with the cue on the Y-axis. The matrix reflects the frequency with which a label was added to an existing sample.

The reformulation into a multi-label classification problem is illustrated in Figure 3.1. Notably, some labels exhibit hierarchical or asymmetric relationships. For instance, the

label *Yawn* is always accompanied by the label *Mouth Open*, but not the reverse. Similarly, *Hand-to-Face* is always present when *Hand-to-Mouth* is annotated. These one-sided relationships suggest a form of behavioral entailment, where the presence of one label implies another. These conditional dependencies are visualized in the entailment diagram in Figure 3.2.

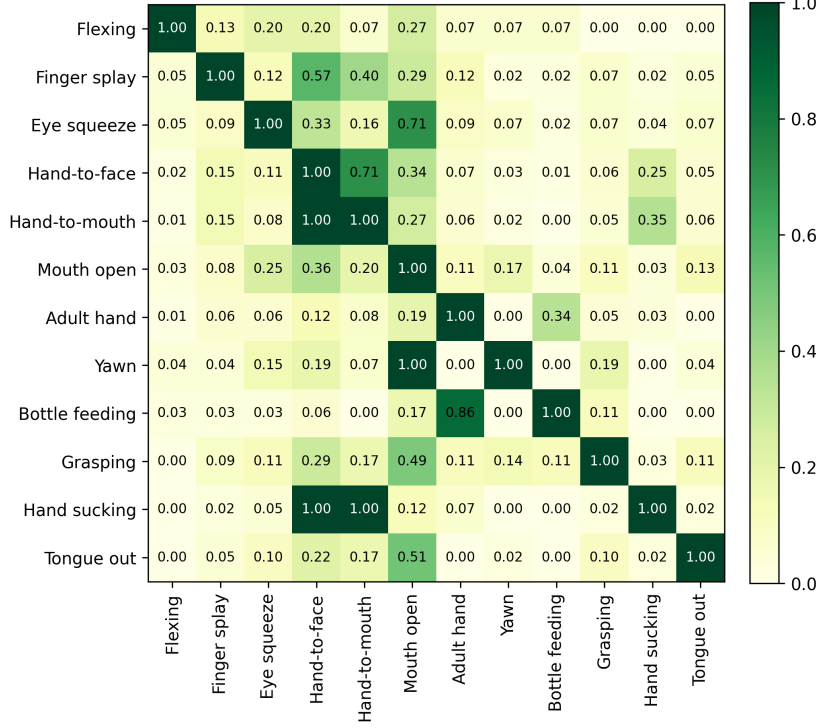


Figure 3.2: Entailment matrix $P(B|A)$ derived from multi-label annotated SLAPI data. The X-axis denotes antecedent labels A , and the Y-axis denotes consequent labels B . Each cell reflects the empirical conditional probability that label B is present given that label A is present

3.1.3 Cross-Validation

The dataset exhibits a strong class imbalance, with the number of annotations ranging from 15 to 164 across classes. Moreover, frames that originate from the video of infant 4 are strongly underrepresented in the data, with only 25 included frames covering 6 classes. To account for variability in label distributions across videos and to reduce the risk of evaluation bias due to overfitting, we employ 5-fold cross-validation. This approach allows for a more reliable estimation of the model’s performance across varying subsets of the data and helps ensure that the evaluation reflects generalization across both common and rare classes.

Several considerations guided the selection of the cross-validation folds. Firstly, it is essential that each fold approximates the overall label distribution of the dataset. This is

important to prevent biased evaluation outcomes: if certain classes are underrepresented or entirely missing in a fold, the model’s performance on those classes can’t reliably be assessed. The rarest class label in the dataset, *Flexing*, occurs a total of 15 times. In an ideally distributed scenario, this would result in three instances of this label per fold. However, to accommodate practical constraints, such as multi-label dependencies and subject-specific sampling limitations, we require that each label is represented at least twice in the validation set of every fold, introducing a tolerance margin of one sample for the label *Flexing*.

To generalize this principle for all labels, it is argued that all classes should deviate a maximum amount from their ideal distribution, relative to that of the least occurring class, *Flexing*. Specifically, for a label with C total occurrences, the ideal per-fold count is C/K , where K is the number of folds. We compute the relative deviation in each fold using Equation 3.1:

$$rel_dev = \frac{|\text{count} - \text{ideal}|}{\text{ideal}} \quad (3.1)$$

For the label *Flexing*, with a minimum of two samples per fold, the relative deviation becomes $rel_dev = \frac{|2-3|}{3} = 0.33$. This value is adopted as an upper bound for the other labels. That is, in each fold, no label should deviate from its ideal distribution by more than 33%. This constraint ensures that each validation set remains representative of the full label distribution.

Secondly, in addition to maintaining proper label distribution across folds, it is important to minimize the separation of video samples originating from the same video across training and validation sets. Annotated frames extracted from the same video are inherently more similar to each other than to frames from other videos, both in terms of visual appearance and contextual cues. In particular, annotated frames from the same video that are temporally adjacent are likely to contain overlapping labels due to continuity in infant behavior. Distributing these frames across folds introduces the risk of information leakage, whereby the model inadvertently learns from validation-like samples during training, thereby compromising the validity of the evaluation.

To minimize this risk, frames originating from the same video should be clustered within a single fold as much as possible. We quantify the effectiveness of this clustering by defining a *GroupingScore*, which measures how well each group (i.e. video) is kept intact between the train and validation sets. The score ranges from 0 to 1. A score of 1 indicates that groups are perfectly separated between train and validation sets in each fold, whereas a score of 0 indicates that the frames from each video are perfectly distributed across all folds. For K folds and a set of groups G . The grouping score is defined in Equations 3.2 and 3.3:

$$GroupingScore = \frac{1}{K|G|} \sum_{k=1}^K \sum_{g \in G} |p_{train}(g) - p_{val}(g)| \quad (3.2)$$

Where:

$$p_{train}(g) = \frac{n_{train}(g)}{n_{train}(g) + n_{val}(g)} \quad p_{val}(g) = \frac{n_{val}(g)}{n_{train}(g) + n_{val}(g)} \quad (3.3)$$

$$n_{train}(g) = \#\{\text{train samples in group } g\}$$

$$n_{val}(g) = \#\{\text{val samples in group } g\}$$

To generate stratified and group-aware folds, an algorithm derived from the *Iterative Multi-Label Stratification* algorithm as proposed by [Sechidis et al. \[2011\]](#) is implemented. The Iterative Multi-Label Stratification algorithm splits multi-label datasets into folds while preserving the global label distribution. For each label, it computes how often it occurs, and then distributes samples to the folds iteratively, starting with samples containing the least frequently appearing label. For each label, the algorithm computes the current deficiency across folds, and assigns the next sample to the fold with the highest label deficiency (e.g. folds that currently have the fewest samples for that label). This approach helps ensure that each fold approximates the same proportion of each label in the dataset.

However, the original Iterative Multi-Label Stratification algorithm does not take into account group structure. The clustering of annotated samples is, as discussed earlier, critical to prevent information leakage between training and validation data. Therefore, a modified version of the Iterative multi-label Stratification algorithm is proposed that incorporates a slack parameter ϵ , that functions as a soft grouping constraint. This adaptation introduces a trade-off between label stratification and grouping. In practice, candidate folds are selected from those whose label deficiency D_l for the current label satisfies Equation 3.4. Where $D_{l,max}$ refers to the maximum label deficiency across all folds for the current label l .

$$\text{Candidates} = \{\text{folds } k | D_l(k) \geq D_{l,max} - \epsilon\} \quad (3.4)$$

Within this candidate set, ties are broken by preferring folds that already contain more samples from the current group, thereby promoting group cohesion. The parameter ϵ thus acts as a soft constraint that allows some tolerance in stratification to gain improvements in group separation. The complete algorithm is shown in Appendix A.

We evaluate the impact of ϵ on both the label stratification quality and the grouping score. Specifically, ϵ is increased from 0 (no tolerance) up to larger values. For $\epsilon = 2$ we find the highest grouping value whilst satisfying the previously defined stratification constraint $rel_dev \leq 0.33$ for all labels across all folds. As shown in Table 3.4, increasing ϵ improves group cohesion at the cost of greater deviation from ideal label proportions.

ϵ	Grouping Score	rel_dev_{max}
0	0.601	0.11
1	0.638	0.33
2	0.676	0.33
3	0.726	0.44

Table 3.4: Trade-off between *GroupingScore* and rel_dev_{max} for different values of ϵ . For $\epsilon = 3$, $rel_dev \leq 0.33$ is no longer satisfied, with *Hand Sucking* (fold 1: $0.39 > 0.33$) and *Yawn* (fold 3: $0.44 > 0.34$) exceeding relative deviation criteria. The setting $\epsilon = 2$ offers a principled trade-off, satisfying the stratification constraints while reducing information leakage.

3.2 Model Selection

For the experiments conducted in this thesis, we select the **Qwen2.5-VL-3B-Instruct** model [Wang et al., 2024b], an auto-regressive decoder-only VLM described in Section 2.3.1. Several factors motivate the decision to choose this model.

Firstly, given the inherent data scarcity in the NICU domain, it is essential to use a model that demonstrates strong task comprehension in zero-shot scenarios. Models that generalize effectively with limited or no task-specific training data are particularly beneficial in this context. Qwen2.5-VL is instruction-tuned, which enhances its capability to better understand and perform tasks specified through natural language prompts. Instruction tuning involves fine-tuning the model on a diverse range of tasks expressed explicitly through human-readable prompts [Liu et al., 2024]. This encourages the model to develop generalized reasoning patterns and a robust understanding of task semantics, making the model better equipped to interpret and execute novel tasks even without task-specific training data.

Additionally, since further research could expand into video understanding, candidate VLMs include those that support video processing. This allows future extensions of this research to incorporate temporal data without requiring a fundamental change in the architecture. Qwen2.5-VL satisfies this criterion because its vision encoder is explicitly designed to handle both image and video modalities. This characteristic ensures that future developments involving temporal data analysis can leverage the existing model architecture without any large architectural modifications.

Finally, the chosen model variant demonstrates strong performance on general visual question answering benchmarks, reinforcing its suitability as a robust baseline for evaluating neonatal behavioral cues.

Practical considerations such as computational resources influence more fine-grained model selection. There are several variably sized versions of the Qwen2.5-VL architecture, as depicted in Table 3.5. Throughout all versions, the Vision Transformer architecture is identical. The parameter size differences are primarily present in the LLM component. The experiments will be executed on high-performance computing (HPC) servers at UMC Utrecht, which possess sufficient capacity to handle the 7B model variant. However, selecting a smaller sized model like the Qwen2.5-VL-3B-Instruct ensures feasibility for potential deployment on edge devices, allowing for real-world application. Collectively, these considerations confirm Qwen2.5-VL-3B-Instruct as a suitable model choice for the objectives of this thesis.

Configuration	Qwen2.5-VL-3B	Qwen2.5-VL-7B	Qwen2.5-VL-72B
Vision Transformer (ViT)			
Hidden Size	1280	1280	1280
# Layers	32	32	32
# Heads	16	16	16
Intermediate Size	3456	3456	3456
Patch Size	14	14	14
Window Size	112	112	112
Full Attention Block Indices	{7, 15, 23, 31}	{7, 15, 23, 31}	{7, 15, 23, 31}
Vision-Language Merger			
In Channel	1280	1280	1280
Out Channel	2048	3584	8192
Large Language Model (LLM)			
Hidden Size	2048	3584	8192
# Layers	36	28	80
# KV Heads	2	4	8
Head Size	128	128	128
Intermediate Size	4864	18944	29568
Embedding Tying	✓	✗	✗
Vocabulary Size	151646	151646	151646
# Trained Tokens	4.1T	4.1T	4.1T

Table 3.5: Model configuration details for Qwen2.5-VL variants, as outlined in the Qwen2.5-VL technical report [Bai et al., 2025]

4 Experimental Design and Results

This section presents a series of experiments designed to evaluate and improve the applicability of VLMs for multi-label cue classification of NICU video data. Each experiment builds upon the findings of the previous one: starting with an exploration of the VLM’s internal embedding space, followed by a text-based classification pipeline based on generated responses, and concluding with a prompt ensembling approach to address robustness. For each experiment, we detail the motivation, methodology, and results separately.

4.1 Embedding Space Visualization

The objective of the first experiment is to assess the feasibility of using VLMs for cue classification. Specifically, we aim to evaluate whether the model can capture and structure semantic information about visual cues in its embedding space, depending on the form and presence of textual instructions.

We hypothesize that a single forward pass through the model should yield a final-layer embedding that encodes semantic representations of both the input image and the associated question prompt. If the model possesses sufficient semantic awareness, then samples that are semantically similar (i.e., that share similar cue labels) should be located near each other in the resulting embedding space. Furthermore, we expect that this semantic space can be modulated through prompt engineering, meaning that textual instructions can influence which features the model attends to and encodes.

4.1.1 Methodology

We compare model embeddings under different prompt conditions: (1) *no textual prompt*, (2) *open-ended questions*, and (3) *binary closed questions* targeting the presence of a specific cue. If the VLM can be effectively guided by the textual prompt, then samples sharing the same label (e.g., presence of a specific cue) should form more coherent clusters in embedding space when prompted than when left unprompted. Contrarily, embeddings generated without prompts are expected to cluster primarily based on visual similarities, such as temporal or contextual proximity within the same video. As such, samples from the same video are expected to be closer together in embedding space when there is no textual prompt.

We hypothesize that open-ended prompts, while semantically rich, may produce overly general embeddings that fail to differentiate between fine-grained cues. This is especially challenging given the presence of multiple co-occurring labels across samples, which

increases semantic variability and makes label-wise clustering more difficult. By contrast, closed binary questions explicitly direct the model to assess whether a specific behavioral cue is present or absent, thereby simplifying the classification task into two distinct categories. Consequently, we expect closed questions to yield more discriminative embeddings between positive and negative samples of a given cue.

The open-ended questions used for this experiment are shown in Table 4.1 and the closed questions in Table 4.2. To analyze the resulting embedding structures, we apply t-distributed stochastic neighbor embedding (t-SNE) [Van der Maaten and Hinton, 2008], a non-linear dimensionality reduction technique that projects high-dimensional embeddings into two or three dimensions. This enables us to visually inspect clustering behavior across different prompting conditions. Plots are colored to highlight various properties in the data, including cue presence, video origin, and temporal ordering of frames.

#	Question
1	<i>Describe the image</i>
2	<i>Give a detailed description of the infant's behavior based on what it's doing with its mouth, hands, and eyes. Make sure to discuss the following labels: {Flexing, Finger Splay, Eye Squeeze, Hand-to-Face, Hand-to-Mouth, Mouth Open, Adult Hand, Yawn, Bottle Feeding, Grasping, Hand Sucking, Tongue Out}</i>
3	<i>Fill in: Which actions is the infant performing?, choose from the following options: {Flexing, Finger Splay, Eye Squeeze, Hand-to-Face, Hand-to-Mouth, Mouth Open, Adult Hand, Yawn, Bottle Feeding, Grasping, Hand Sucking, Tongue Out}</i>
4	<i>Describe the image of the infant based on the following behavioral attributes: {Flexing, Finger Splaying, Eye Squeezing, Hand-to-Face, Hand-to-Mouth, Mouth Open, Adult Hand, Yawn, Bottle Feeding, Grasping, Hand Sucking, Tongue Out}</i>
5	<i>Fill in: which of the following actions apply to the infant?, choose from the following options: {Flexing, Finger Splay, Eye Squeeze, Hand-to-Face, Hand-to-Mouth, Mouth Open, Adult Hand, Yawn, Bottle Feeding, Grasping, Hand Sucking, Tongue Out}</i>

Table 4.1: List of the open-ended question prompts submitted to the model alongside the frame of the annotated sample, each question aims to retrieve relevant textual information of all labels

#	Label	Question
1	Flexing	<i>Is the infant flexing its body?</i>
2	Finger Splay	<i>Is the infant splaying its fingers?</i>
3	Eye Squeeze	<i>Is the infant squeezing its eyes?</i>
4	Hand-to-Face	<i>Is the infant's hand touching its face?</i>
5	Hand-to-Mouth	<i>Is the infant's hand touching its mouth?</i>
6	Mouth Open	<i>Is the infant's mouth open wide?</i>
7	Adult Hand	<i>Is there an adult hand visible in the image?</i>
8	Yawn	<i>Is the infant yawning?</i>
9	Bottle Feeding	<i>Is the infant being bottle fed?</i>
10	Grasping	<i>Is the infant grasping (onto something with) its hand?</i>
11	Hand Sucking	<i>Is the infant sucking on its fingers or the back of its hand?</i>
12	Tongue Out	<i>Is the infant sticking its tongue out?</i>

Table 4.2: List of the closed question prompts submitted to the model alongside the frame of the annotated sample. Each question aims to assess the presence of a single class.

4.1.2 Results

In the embedding space visualization experiment, the multilabel annotated dataset is used. During inference, four samples failed to process ($\{sample=150, infant=2, cues=[Hand-to-Face, Hand-to-Mouth]\}$, $\{sample=159, infant=2, cues = [Hand-to-Face, Hand-to-Mouth, Mouth Open]\}$, $\{sample=306, infant=3, cues=[Hand-to-Face, Mouth Open]\}$, $\{sample=318, infant=3, cues=[Hand-to-Face, Hand-to-Mouth, Adult Hand]\}$). Figure 4.1 shows the resulting t-SNE embeddings for all other samples when no textual instructions are provided. Each point corresponds to a single video frame, and colors denote the video of origin. As expected, embeddings form tight clusters based on their video of origin, indicating that in the absence of textual input, the model primarily encodes visual similarities rather than cue-specific semantic distinctions.

In Figure 4.2, t-SNE embeddings for infants 1 through 4 are presented separately. Infant 5 is excluded from these individual plots due to an insufficient number of annotated samples, which prevents the reliable formation of visible clusters. In these visualizations, color presents the temporal order of the frames within each video. Embeddings which are temporally adjacent in the video are also assigned adjacent colors. Across all four infants, we observe that temporally neighboring frames tend to cluster closely in the embedding space, suggesting that the VLM encodes visual similarities when no textual prompt is provided. Notably, more distinct and diverse color clusters are visible in Figures 4.2a and 4.2b, which may correspond to repeated behaviors within the video.

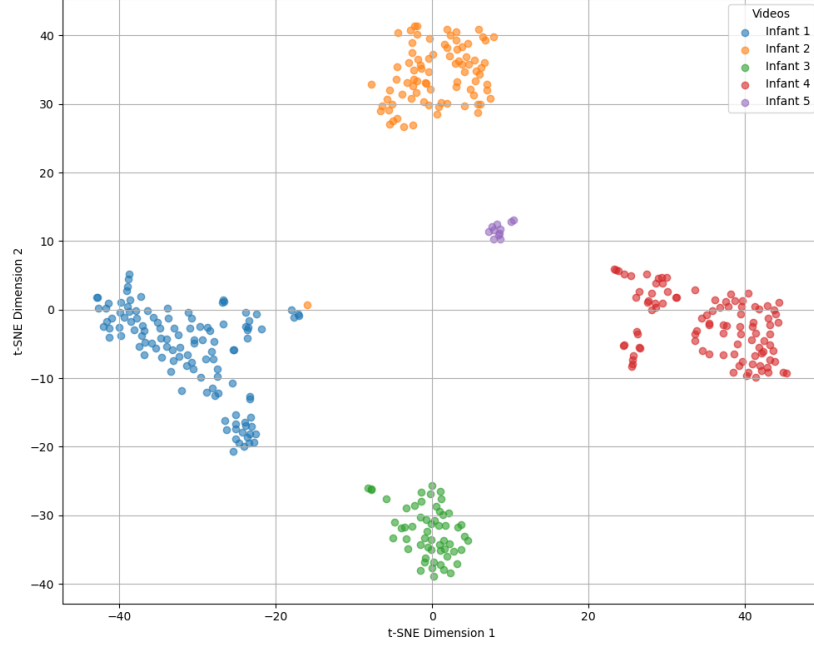
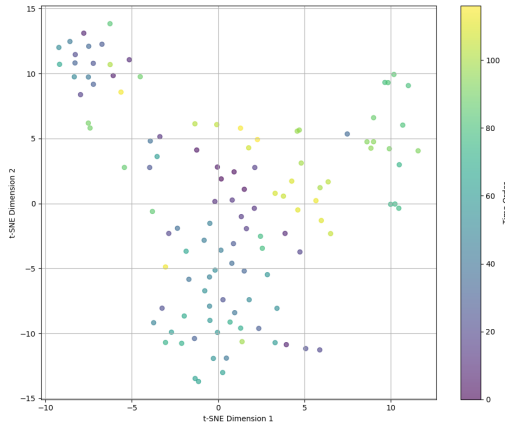
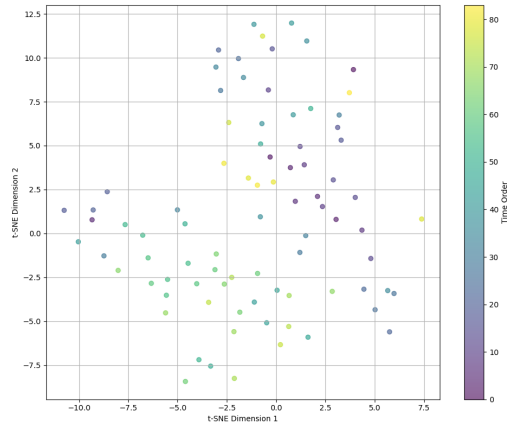


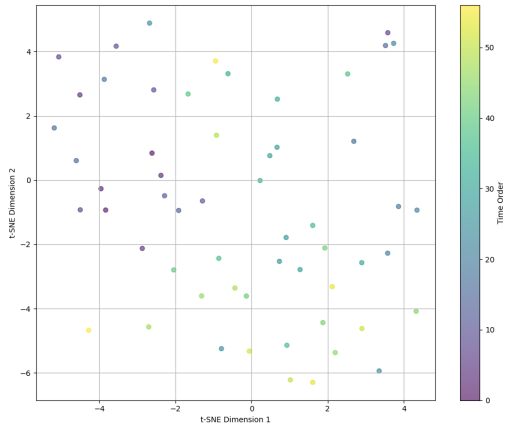
Figure 4.1: t-SNE embeddings of annotated samples across all recorded infants. The embeddings are obtained by forwarding each sample frame through the VLM without providing any textual prompt. The samples are colored according to their video of origin.



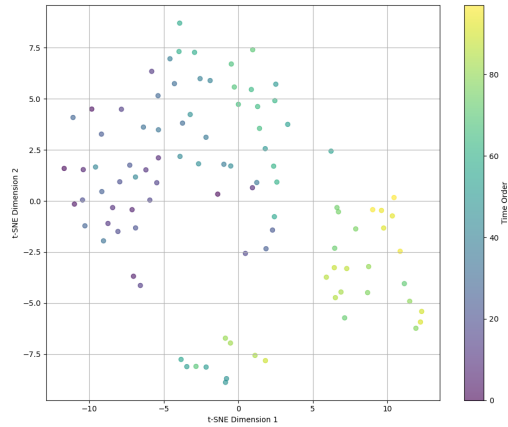
(a) Infant 1



(b) Infant 2



(c) Infant 3



(d) Infant 4

Figure 4.2: t-SNE embeddings of annotated samples from infants 1-4. The embeddings are obtained by forwarding each sample frame through the VLM without providing any textual prompt. The samples are colored according to their time order.

Some of these clusters appear to reflect consistent visual patterns associated with specific cues. In particular, the model seems to attend to the cues *Adult Hand* and *Bottle Feeding*, even without textual guidance. This is further illustrated in Figures 4.3 and 4.4. These cues frequently co-occur in the dataset, with a conditional probability of $P(\text{Bottle Feeding} \mid \text{Adult Hand}) = 0.86$ (see Figure 3.2), which likely contributes to their overlapping representation in embedding space. A full overview of cue-wise label presence in the no-prompt condition is provided in Appendix B, Figure B.1.

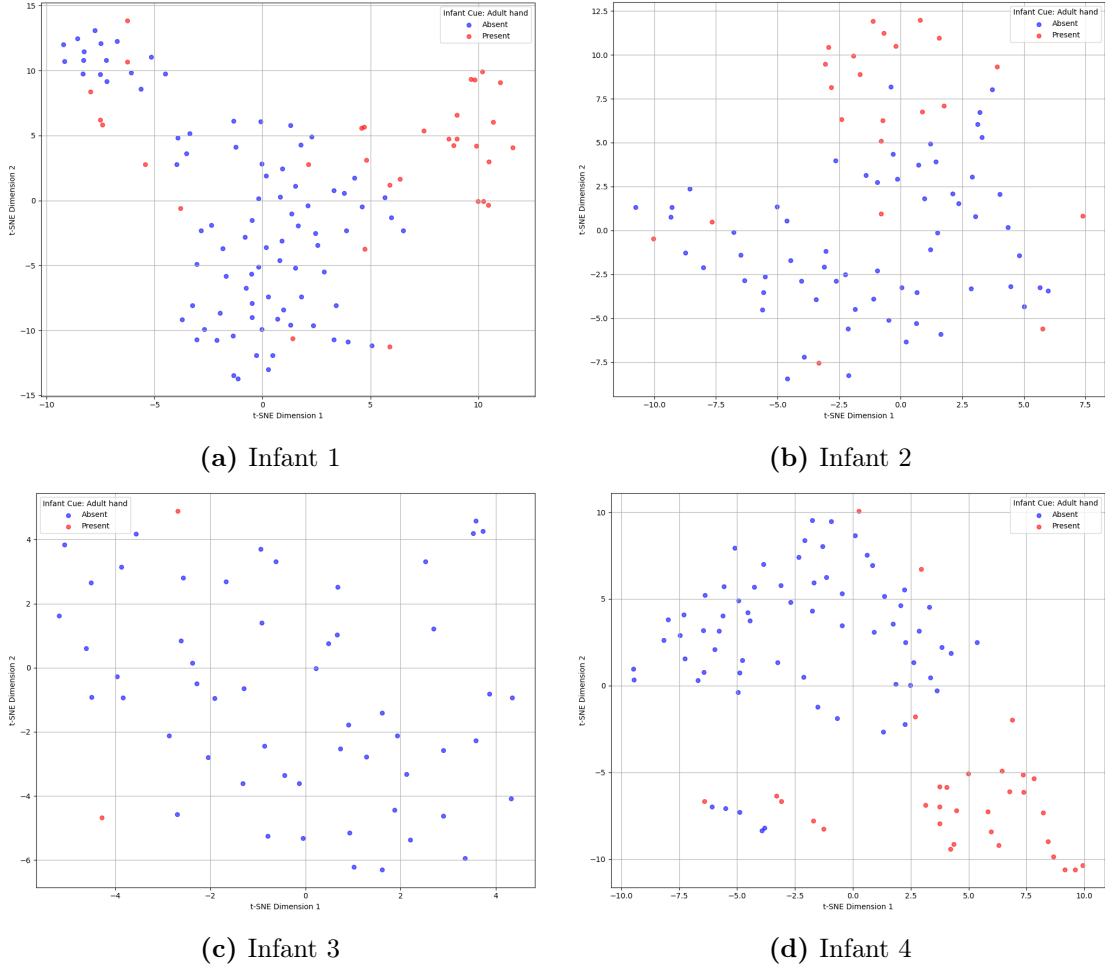


Figure 4.3: t-SNE embeddings of annotated samples from infants 1-4. The embeddings are obtained by forwarding each sample frame through the VLM without providing any textual prompt. Points are colored based on the presence of the label *Adult Hand*: red indicates presence, and blue indicates absence.

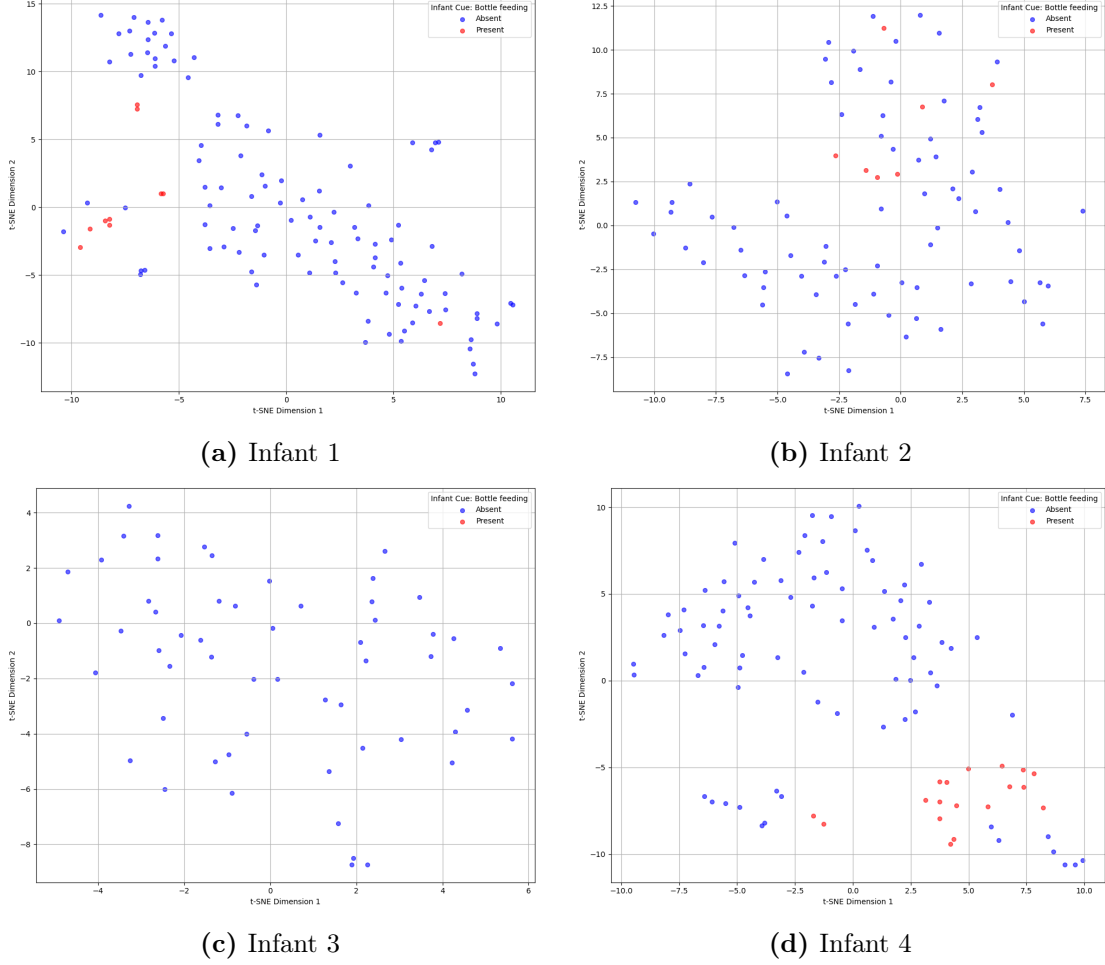


Figure 4.4: t-SNE embeddings of annotated samples from infants 1-4. The embeddings are obtained by forwarding each sample frame through the VLM without providing any textual prompt. Points are colored based on the presence of the label *Bottle Feeding*: red indicates presence, and blue indicates absence.

The same experiment was repeated using both open-ended and closed-question prompts. Figure 4.5 displays example embeddings generated from open-ended question #4 (see Table 4.1). In Figure 4.5a, we observe that embeddings from different infant videos exhibit increased proximity compared to the no-prompt condition shown in Figure 4.1. In particular, embeddings from infants 1 and 4 appear to attract one another in the reduced embedding space. However, despite this cross-group attraction, we do not observe distinct clusters corresponding to specific cues across videos. Among the merged embeddings, the cue most frequently present is *Adult Hand*, as visualized in Figure 4.5b.

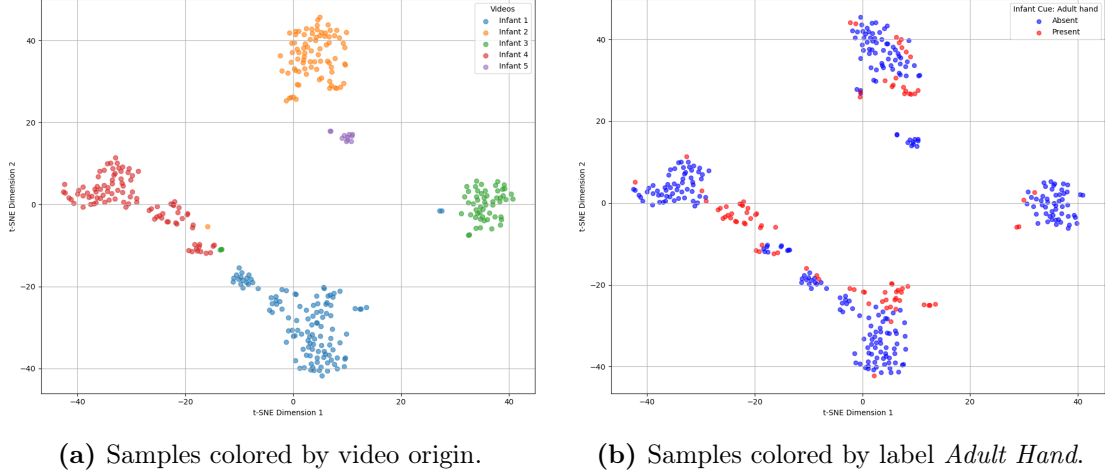


Figure 4.5: t-SNE embeddings of annotated samples across all infants. The embeddings are obtained by forwarding each sample frame and open-ended question #4 (see Table 4.1) through the VLM. The samples are colored based on video origin (4.5a) and whether label *Adult Hand* is present (red) or not (blue) (4.5b). Samples between infant 1 and 4 show convergence in embedding space, with *Adult Hand* being the most prevalent label within this region.

Closed-questions generally yield similar patterns to those observed with open-ended prompts, showing limited cue-specific clustering across infant videos. However, an exception is observed for the cue *Bottle Feeding*, associated with closed question #9 (see Table 4.2). In this case, embeddings for infants 1 and 4 form a more distinct cluster, particularly around the presence of the *Bottle Feeding* cue. This suggests that, in some cases, closed questions can guide the model toward generating more discriminative embeddings. We hypothesize that this primarily occurs for cues that are visually salient, meaning the cues dominate the visual scene or involve distinct objects or interactions (e.g. a bottle or an adult hand). The corresponding t-SNE plots are shown in Figure 4.6. A full overview of cue-wise label presence using closed prompts is provided in Figure B.2, Appendix B.

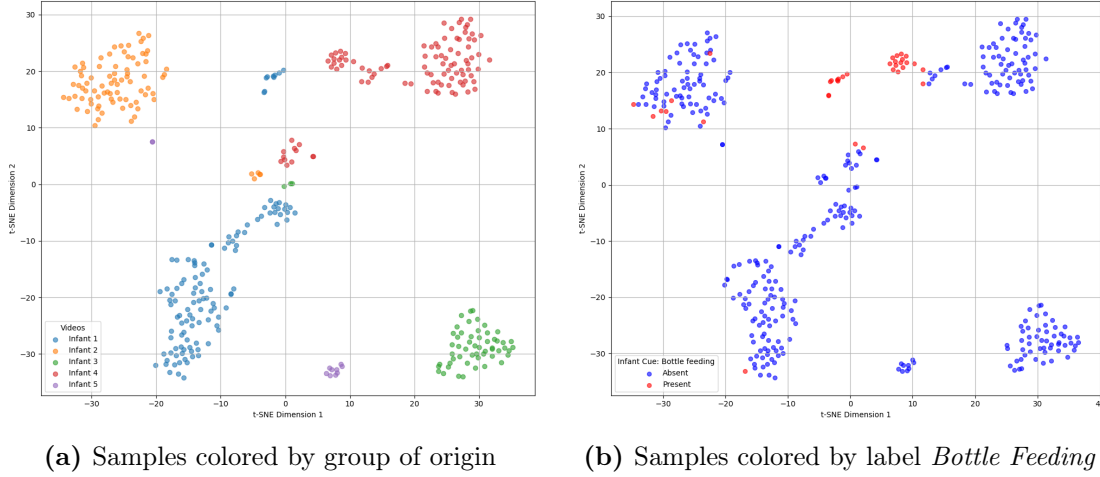


Figure 4.6: t-SNE embeddings of annotated samples across all infants. The embeddings are obtained by forwarding each sample frame and closed question #9 (see Table 4.2) through the VLM. The samples are colored based on video origin (4.6a) and whether label *Bottle Feeding* is present (Red) or not (Blue) (4.6b). Samples between infant 1 and 4 exhibit clear clustering.

When looking at embeddings within videos, some change is observed when using closed-question prompts. Specifically, cue-related clusters that are already visible without prompting appear to become more distinct. This distinction between present and absent samples is most noticeable when closed-question prompts directed towards *Bottle Feeding* and *Adult Hand* are used. An example is shown in Figure 4.7, which compares embeddings of infant 1 under three prompting conditions: no textual prompt, an open-ended question, and a closed question targeting *Adult Hand*. While all three conditions reveal some degree of structure, the closed-question condition leads to the clearest separation between positive (red) and negative (blue) samples.

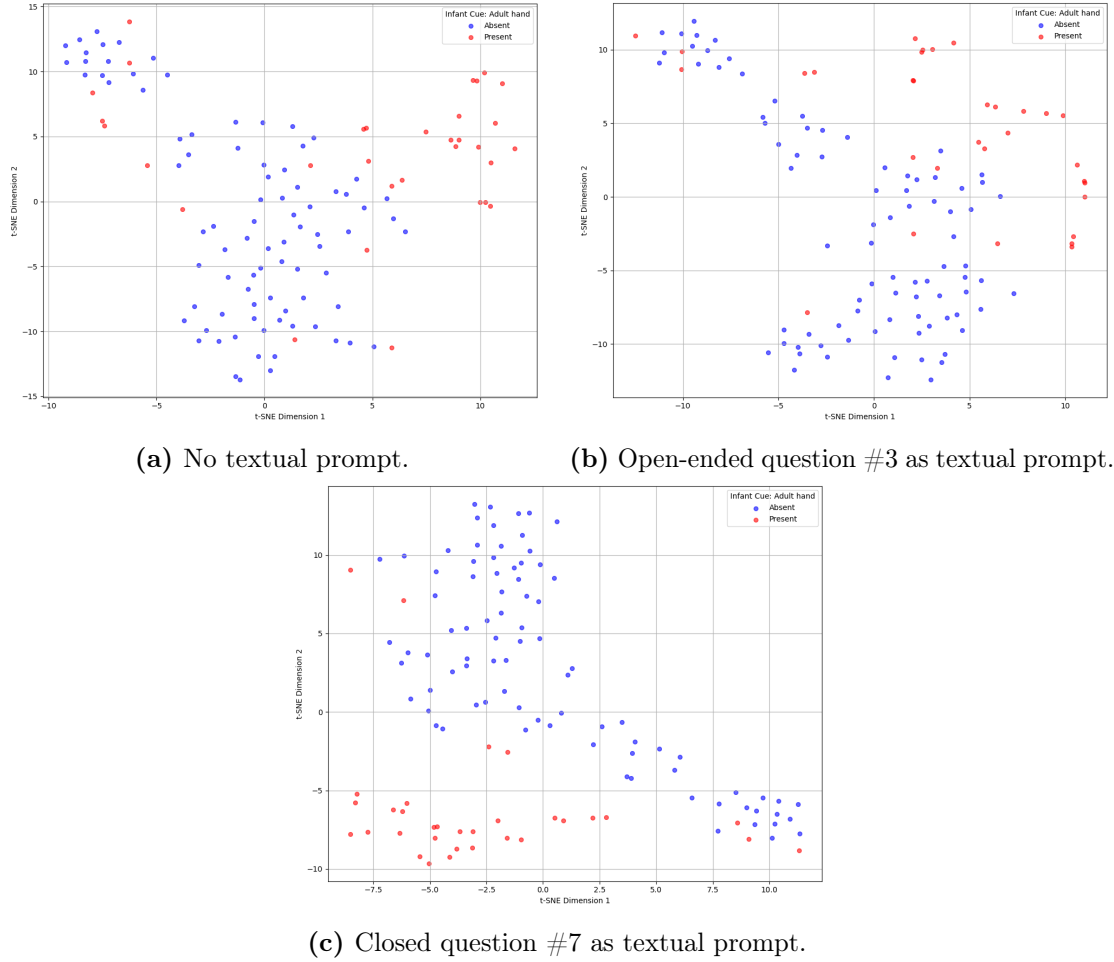


Figure 4.7: t-SNE embeddings of annotated samples of infant 1. The embeddings are obtained by forwarding each sample frame combined with different prompt compositions through the VLM. Open-ended and closed question prompts are defined in Tables 4.1 and 4.2. Samples are colored based on whether label *Adult Hand* is present (Red) or not (Blue). Embeddings generated using a closed question prompt exhibit the clearest separation.

Other behavioral cues did not exhibit meaningful clustering in the embedding space, either between or within videos. Moreover, these cues did not show any substantial improvement when using closed-question prompts. An example of this is shown in Figure 4.8, which visualizes the embeddings associated with the label *Finger Splay* across different prompting conditions. No clear separation emerges under any of the three prompting strategies. While a small degree of local grouping can be observed in some regions under the closed-question prompt, these patterns are weak and possibly subjective. This suggests textual prompting does not substantially improve the model’s embedding structure for cues that are less visually salient. An alternative explanation is that the model has not been sufficiently exposed to these cues during pretraining, limiting its capacity to distinguish them in the provided frames.

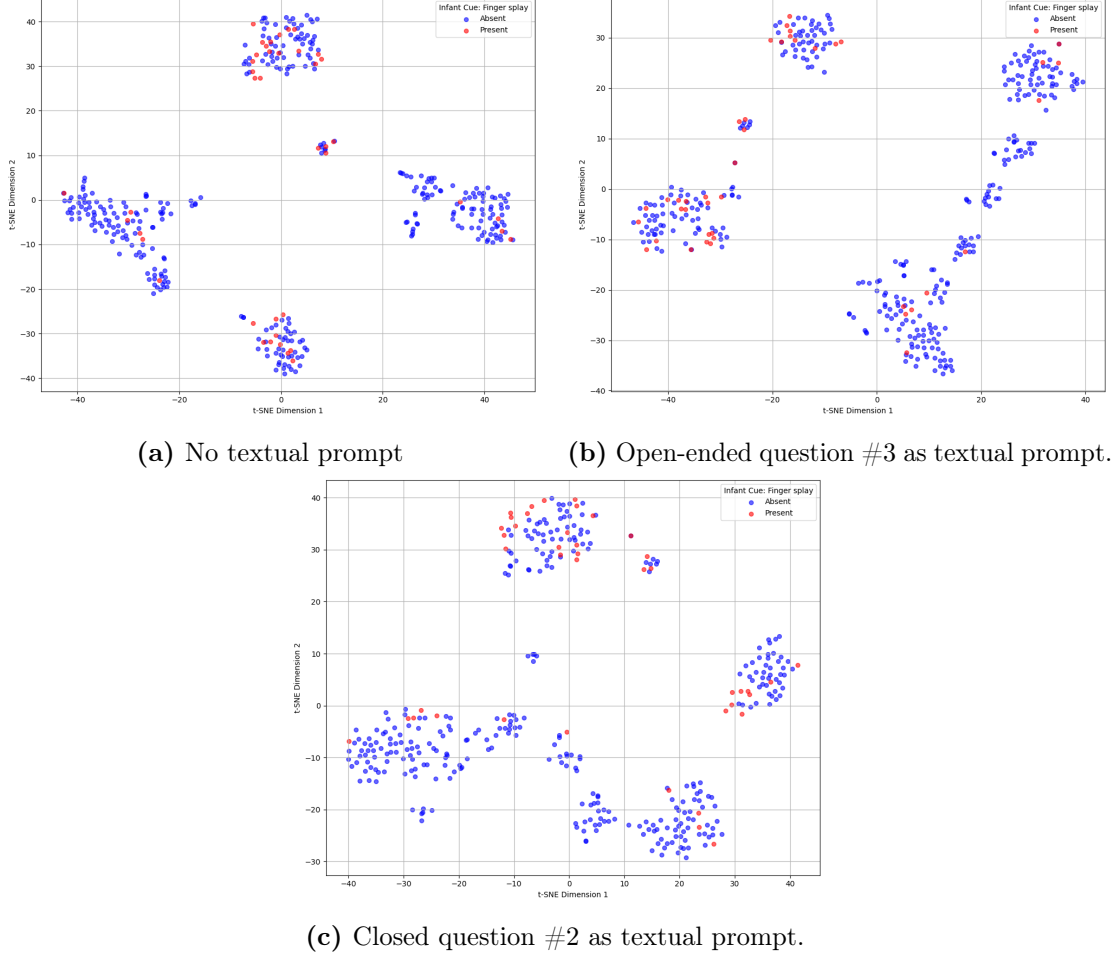


Figure 4.8: t-SNE embeddings of annotated samples across all infants. The embeddings are obtained by forwarding each sample frame combined with different prompt compositions through the VLM. Open-ended and closed question prompts are defined in Tables 4.1 and 4.2. Samples are colored based on whether label *Finger Splay* is present (Red) or not (Blue). Embeddings generated using a closed question prompt exhibit little to no clustering. A small amount of clustering is observed in some groups in 4.8c

4.2 Text-based Classification

The second experiment builds on the insights from the embedding space visualization. In the previous experiment, we examined the VLM’s final-layer embeddings to assess whether samples with similar labels formed coherent clusters. While the use of textual prompts improved alignment to some degree, we observed that the embeddings predominantly clustered according to the video of origin. This suggests that the representations are influenced more by scene-level similarities that have likely propagated through the model by the ViT, rather than by cue-related features.

Secondly, although embedding visualization offers interpretability, it does not facilitate direct classification. To enable discrete classification, We identify two possible approaches:

1. By training a classification head on top of the final layer’s embeddings
2. By performing text-based classification by comparing the decoder-generated output to pre-defined textual templates.

The first approach poses two key limitations. First, training a classification-head requires a substantial amount of data, which is scarce within the NICU domain. Second, given the poor generalization of embeddings across videos, a classification head is unlikely to overcome this problem. Instead, it is likely the model will overfit on the currently available data, and will fail to classify samples from novel videos correctly.

The second approach addresses these challenges. By performing classification based on the model’s textual output, we can decouple text-generated predictions from the visual context. This may potentially increase the generalization capabilities of the model. Importantly, this method does not require any additional model training. The only tunable parameter is a similarity threshold that determines whether a generated sentence sufficiently aligns with a predefined template. This lightweight and interpretable approach makes it well-suited for data-scarce domains.

In the previous experiment, closed prompts did improve class separation for a few highly prevalent cues (e.g., Bottle Feeding, Adult Hand), but these improvements were not consistent across more subtle cues. This suggests that prompting alone is insufficient to make all cues distinguishable in embedding space. By instead evaluating the model’s textual output, we gain a more interpretable measure of the model’s understanding of both the image and the associated prompt. This is favorable in healthcare settings where transparency is highly valued.

4.2.1 Methodology

For each behavioral cue, we construct a closed-question prompt alongside a corresponding template answer that represents the presence of that cue in the frame. After generation, the model’s output text will be compared to the template by computing the cosine-similarity between their embeddings. An overview of the prompt and template pairs used in this experiment is presented in Table C.1, Appendix C. Importantly, we compute the embedding from a forward pass of the generated texts themselves, rather than from the multimodal input. This decouples the visual context from the text-generated predictions. The embedding used for computing the cosine-similarity is the hidden state from the final transformer layer. This token’s hidden state is assumed to encode contextual information from the entire sequence, as it attends to all previous tokens using self-attention [Vaswani, 2017].

To determine whether a cue is present in a given sample, we threshold the cosine similarity

between the generated output and the corresponding template. If the cosine-similarity of a label exceeds the threshold, the label will be rendered present. Thresholds are computed in separate training folds, using the cross-validation splits described in Section 3.1.3. We evaluate two thresholding strategies: (1) a *local threshold* that is distinct for each label, and (2) a *global threshold* that is shared across all labels. Thresholds are selected by optimizing the F_1 -score on the train-set. In order to select global thresholds, the F_1 -score is micro-averaged. It is hypothesized that local thresholds will yield a higher classification performance, as the average cosine-similarity score may vary across different question-answer pairs.

One limitation the suggested approach may pose, is that the VLM is not explicitly trained to distinguish between positive and negative responses. We expect that the embeddings derived from the forward pass may embed semantically opposite answers in similar regions of the embedding space. This may result in minimal differences in cosine similarity scores between generated outputs and their corresponding templates, increasing the risk of incorrect classification. To address this, we evaluate several alterations.

First, we hypothesize that including the question as context alongside the generated answer may produce richer textual embeddings, resulting in better discrimination between texts that describe a label as present and texts that describe a label as absent. To evaluate this, we will compare the answer templates of composition $\{Expected\ Answer\}$ to templates of composition $\{Question\ [SEP]\ Expected\ Answer\}$, where $[SEP]$ represents a separation token used to disambiguate the question and answer components.

Second, we will compare embeddings created by the VLM with those from a specialized sentence transformer trained to cluster semantically similar and dissimilar texts. We hypothesize that using a dedicated model will yield more discriminative representations and create larger gaps between positive and negative outputs. Specifically, we will use `all-MiniLM-L6-v2`: a distilled sentence transformer derived from Microsoft’s MiniLM-L12-H384 Wang et al. [2020]. The model is fine-tuned using contrastive learning and is intended for information retrieval, clustering, and sentence similarity tasks. This makes it a suitable model to discriminate positive and negative sentences in embedding space.

We compare the performance of the proposed approaches against three simple yet informative baseline classifiers.

- **Random classifier:** Predicts each label as present with a fixed probability:

$$P(\text{Label}) = 0.5$$

This baseline reflects uninformed guessing and establishes a theoretical lower bound

for model performance.

- **Majority classifier:** Predicts only the most frequently occurring label in the training set as present for all samples:

$$P(\text{Label}) = \begin{cases} 1 & \text{if label} = \arg \max_{l \in \mathcal{L}} \text{count}(l) \\ 0 & \text{otherwise} \end{cases}$$

This classifier captures dataset sampling bias towards the dominant class while completely ignoring minority labels.

- **Prior-based classifier:** Predicts each label as present with probability proportional to its frequency in the training set:

$$P(\text{label}) = \frac{\text{count}(\text{label})}{N}$$

where $\text{count}(\text{label})$ is the number of times the label occurs in the training set, and N is the total number of label occurrences across all samples. This baseline offers an “educated guess” that accounts for class imbalance, and is expected to perform slightly better than random guessing.

Performance is evaluated using precision, recall and F_1 -score. Each of these metrics is reported with both macro- and micro-averaging. Macro-averaging computes metrics independently for each label and then averages across labels, weighing each class evenly. Micro-averaging averages over all labels across all samples, weighing each sample evenly. In the context of NICU cue classification, where the clinical goal is to assess the infants behavioral or physiological state (e.g. sleep, pain, hunger), all types of cues are equally important to make this assessment. Therefore, we consider macro-F1 to be the primary metric for optimization.

Additionally, we report the Jaccard Similarity, Hamming Loss and Subset Accuracy for complementary insights. Jaccard similarity captures the overlap between predicted and ground-truth sets. This metric is valuable because ultimately we want to identify the correct combination of cues rather than individual labels. As different combinations can lead to different assessments of the infants state. Hamming Loss computes the fraction of false positives and false negatives over the total number of predictions, penalizing each incorrect label independently. Lastly, subset accuracy represents the proportion of samples for which all labels are predicted exactly. It is a strict metric and can be seen as an extreme case of Jaccard similarity, requiring full set-level correctness.

An overview of how all metrics are computed is provided in Table [4.3](#)

Metric	Type	Formula
Precision	Micro	$\frac{\sum_{i=1}^N \text{TP}_i}{\sum_{i=1}^N (\text{TP}_i + \text{FP}_i)}$
Precision	Macro	$\frac{1}{L} \sum_{l=1}^L \frac{\text{TP}_l}{\text{TP}_l + \text{FP}_l}$
Recall	Micro	$\frac{\sum_{i=1}^N \text{TP}_i}{\sum_{i=1}^N (\text{TP}_i + \text{FN}_i)}$
Recall	Macro	$\frac{1}{L} \sum_{l=1}^L \frac{\text{TP}_l}{\text{TP}_l + \text{FN}_l}$
F ₁ Score	Micro	$2 \frac{\text{Precision}_\mu \cdot \text{Recall}_\mu}{\text{Precision}_\mu + \text{Recall}_\mu}$
F ₁ Score	Macro	$\frac{1}{L} \sum_{l=1}^L 2 \frac{\text{Precision}_l \cdot \text{Recall}_l}{\text{Precision}_l + \text{Recall}_l}$
Jaccard	Micro	$\frac{\sum_{i=1}^N Y_i \cap \hat{Y}_i }{\sum_{i=1}^N Y_i \cup \hat{Y}_i }$
Jaccard	Macro	$\frac{1}{L} \sum_{l=1}^L \frac{\text{TP}_l}{\text{TP}_l + \text{FP}_l + \text{FN}_l}$
Hamming Loss	—	$\frac{1}{NL} \sum_{i=1}^N \sum_{l=1}^L \mathbf{1}[y_{il} \neq \hat{y}_{il}]$
Subset Accuracy	—	$\frac{1}{N} \sum_{i=1}^N \mathbf{1}[Y_i = \hat{Y}_i]$

N Total number of instances.

L Total number of labels. $\text{TP}_i, \text{FP}_i, \text{FN}_i$ True positives, false positives, and false negatives for instance i . $\text{TP}_l, \text{FP}_l, \text{FN}_l$ True positives, false positives, and false negatives for label l (summed over all instances).

Y_i Ground-truth label set for instance i . $\hat{Y}_i$ Predicted label set for instance i .

Table 4.3: Computational overview of evaluation metrics used in the text-based classification experiment.

4.2.2 Results

Table 4.4 summarizes the performance of the three baseline classifiers across five cross-validation folds.

The random classifier, which predicts each label as present with probability 0.5, achieves the highest micro- and macro-recall scores—approximately 0.51 and 0.53, respectively—due to its tendency to over-predict. However, this comes at the cost of low precision, yielding a micro and macro-precision of only 0.19.

The majority classifier, which always predicts the most frequent label in the training set—*Hand-to-Face*, occurring in 164 out of 371 samples (44%), achieves the highest micro-precision at 0.44. However, since it entirely ignores all other labels, its macro-precision drops significantly:

$$\frac{0.44 + 0 + \dots + 0}{12} = 0.04.$$

We can also estimate the Hamming Loss for the majority classifier:

$$\frac{\text{FP} + \text{FN}}{N \times C} = \frac{207 + 650}{371 \times 12} \approx 0.19,$$

where $N = 371$ is the number of samples and $C = 12$ the number of labels. It is important to note that Hamming Loss reflects the per-label error rate and can appear low for conservative strategies. For example, a classifier that predicts all labels as absent achieves a Hamming Loss of 18.4%. The majority classifier consequently achieves the lowest Hamming Loss (0.20) among the baselines by making very few predictions.

Subset accuracy is essentially zero across all baselines. This is expected, as it requires an exact match between predicted and ground-truth label sets, and even one misclassified label causes the entire prediction to be marked incorrect. The highest subset accuracy is observed in the majority classifier (3%), which corresponds to the proportion of samples that contain only the *Hand-to-Face* label.

Although the random, majority, and prior-based classifiers balance precision and recall differently — micro-precision values of 0.19, 0.44, and 0.27, and micro-recall values of 0.51, 0.20, and 0.27, respectively — their micro- F_1 scores converge at 0.27. Micro-Jaccard scores also align at 0.16 across all three baselines. However, macro-averaged metrics show more discriminative differences, with the random and prior-based classifiers outperforming the majority classifier by a considerable margin. These results underscore the importance of macro-averaging in multi-label settings where a large imbalance in class frequency is present.

Metric stability across folds is relatively high: no standard deviation exceeds 0.05. This indicates that all baseline classifiers perform consistently across different training splits. Stable results are especially desired for the prior-based classifier, as this indirectly implies class distributions are well-balanced between training and test sets across folds.

Baseline classifier	Random		Majority		Prior-based	
	Mean	SD	Mean	SD	Mean	SD
Precision (micro)	0.19	0.02	0.44	0.08	0.27	0.02
Precision (macro)	0.19	0.02	0.04	0.01	0.19	0.02
Recall (micro)	0.51	0.02	0.20	0.02	0.27	0.04
Recall (macro)	0.53	0.03	0.08	0.00	0.19	0.04
F_1 (micro)	0.27	0.02	0.27	0.03	0.27	0.02
F_1 (macro)	0.25	0.02	0.05	0.01	0.19	0.02
Jaccard (micro)	0.16	0.01	0.16	0.02	0.16	0.02
Jaccard (macro)	0.15	0.01	0.04	0.01	0.11	0.02
Hamming loss	0.50	0.01	0.20	0.02	0.27	0.03
Subset accuracy	0.00	0.00	0.03	0.04	0.01	0.01

Table 4.4: Performance of the three baseline classifiers evaluated across five folds, using the metrics defined in Table 4.3. For each metric, the mean and standard deviation (SD) are reported. Bold values indicate the best-performing classifier per metric.

Table 4.5 presents the results of our first VLM-based classification configurations, moving beyond the baselines previously discussed. Specifically, we compare the performance of Qwen2.5-VL across two input compositions (VLM *Answer* only vs. *Question [SEP] Answer*) and two thresholding strategies (global vs. local). All results are means and standard deviations over five cross-validation folds.

Using only the VLM’s generated answer as input outperforms concatenating the original question and answer. This finding suggests that the answer alone encapsulates most relevant semantic information for label prediction, while the inclusion of the question may introduce noise. This rejects our initial hypothesis that adding the initial question as context would result in richer embeddings.

A global threshold yields the highest micro-averaged precision, recall, F_1 , and Jaccard scores. Because micro-metrics are dominated by frequent classes, a global cutoff conservatively avoids over-predicting rare labels, boosting overall (micro) precision and F_1 . In contrast, local thresholds dramatically increase per-class recall, improving macro-averaged recall and F_1 . By lowering thresholds for infrequent classes, the model recovers more true positives on rare labels. This leads to an increase in false positives, thereby explaining the higher Hamming Loss observed with local thresholding.

All Qwen-based configurations outperform the three baseline classifiers across both micro and macro metrics. This indicates that even without explicit fine-tuning, the VLM produces embeddings and answers that capture label-relevant information better than any baseline strategy.

Input composition	<i>Answer</i>				<i>Question [SEP] Answer</i>			
Threshold type	Global		Local		Global		Local	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Precision (micro)	0.37	0.03	0.30	0.03	0.30	0.03	0.28	0.03
Precision (macro)	0.25	0.04	0.32	0.04	0.27	0.03	0.29	0.05
Recall (micro)	0.52	0.04	0.70	0.02	0.62	0.03	0.68	0.03
Recall (macro)	0.38	0.04	0.60	0.05	0.51	0.06	0.57	0.04
F_1 (micro)	0.43	0.03	0.42	0.03	0.41	0.03	0.40	0.02
F_1 (macro)	0.28	0.04	0.37	0.04	0.31	0.03	0.33	0.03
Jaccard (micro)	0.28	0.03	0.26	0.03	0.25	0.02	0.25	0.02
Jaccard (macro)	0.19	0.03	0.26	0.03	0.21	0.02	0.22	0.02
Hamming loss	0.25	0.03	0.36	0.03	0.34	0.03	0.38	0.03
Subset accuracy	0.02	0.02	0.01	0.01	0.00	0.01	0.00	0.00

Table 4.5: Performance of the `Qwen2.5-VL-3B-Instruct` model evaluated across five folds, using the metrics defined in Table 4.3. Four configurations are compared, based on two input compositions (*Answer* vs. *Question [SEP] Answer*) and two thresholding strategies (Global vs. Local). For each metric, the mean and standard deviation (SD) are reported. Bold values indicate the best-performing classifier per metric.

Table 4.6 reports precision, recall, and F_1 (mean $\pm$ standard deviation over five folds) for each of the 12 labels, comparing global vs. local thresholding on the model that uses Answer-only input.

Despite no clear embedding clusters for *Hand-to-Face* and *Hand-to-Mouth* in our visualization experiments, both labels achieve high F_1 scores (0.75 and 0.60, respectively) under both thresholding strategies. This indicates that — even without obvious clustering — the model reliably distinguishes these frequent cues from other classes.

Adult Hand and *Bottle Feeding*, which did show tight clusters in the embedding space, yield poor global-threshold performance ($F_1 = 0.00$) and only modest gains with local thresholds ($F_1 = 0.39$ and 0.51). Their global-threshold performance is a consequence of low optimal local thresholds, as compared to the other classes. Those low optimal thresholds suggest that output embeddings of these classes did not align well with the template queries, resulting in thresholds too low for the global cut-off to capture. This misalignment between generated answers and template answers may lead to indistinguishable embeddings between present and absent labels, resulting in near-identical cosine-similarity scores, which could explain the low performance of these classes.

Another explanation is that the issue here does not reside in the alignment with the answer templates, but within the VLM itself. This could happen either when the closed question input prompt was poorly formulated, or the model does not have a ‘visual understanding’ of the class. From the embedding visualization experiment we have seen that embeddings

for *Bottle Feeding* clustered more when introduced with a closed question prompt. This suggests that the second explanation is unlikely.

Across most classes, local thresholds dramatically increase recall, but precision gains are modest or negative, yielding only small F_1 improvements. This systematic over-prediction suggests that the VLM contains a bias toward presence rather than absence of cues. A possible explanation for this bias is that VLMs are generally trained on image-text pairs where the text usually describes what objects and actions are present in the image, rather than absent ones.

Hand-to-Face, the most frequent label, poses identical metrics between global and local threshold experiments. This is because the global threshold is computed without taking class distribution into account. Consequently, the single global cut-off aligns almost exactly with the Hand-to-Face decision boundary and drives the micro-averaged F_1 in its favor.

Although most individual F_1 scores improve under local thresholds, the micro-averaged F_1 remains slightly higher for the global threshold (0.43 vs. 0.42) because *Hand-to-Face* and *Hand-to-Mouth* dominate the aggregate. In contrast, local thresholds raise rare-class F_1 more substantially, leading to a higher macro-averaged F_1 (0.37 vs. 0.28).

Metric	Global						Local					
	P		R		F_1		P		R		F_1	
	M	SD	M	SD	M	SD	M	SD	M	SD	M	SD
Flexing	0.00	0.00	0.00	0.00	0.00	0.00	0.05	0.06	0.20	0.30	0.07	0.10
Finger splay	0.29	0.21	0.46	0.18	0.33	0.18	0.28	0.23	0.43	0.25	0.32	0.20
Eye squeeze	0.15	0.03	0.30	0.11	0.20	0.04	0.13	0.02	0.67	0.31	0.22	0.04
Hand-to-face	0.72	0.06	0.79	0.06	0.75	0.03	0.72	0.06	0.79	0.06	0.75	0.03
Hand-to-mouth	0.47	0.05	0.83	0.05	0.60	0.05	0.47	0.05	0.83	0.05	0.60	0.05
Mouth open	0.57	0.09	0.69	0.16	0.62	0.12	0.75	0.10	0.59	0.17	0.65	0.11
Adult hand	0.00	0.00	0.00	0.00	0.00	0.00	0.24	0.04	0.99	0.02	0.39	0.05
Yawn	0.37	0.24	0.30	0.15	0.30	0.17	0.38	0.25	0.30	0.15	0.31	0.18
Bottle feeding	0.00	0.00	0.00	0.00	0.00	0.00	0.41	0.24	0.70	0.45	0.51	0.31
Grasping	0.12	0.16	0.06	0.08	0.08	0.10	0.12	0.11	0.42	0.37	0.17	0.14
Hand sucking	0.12	0.05	0.61	0.20	0.20	0.08	0.11	0.03	0.78	0.13	0.19	0.05
Tongue out	0.19	0.07	0.54	0.26	0.28	0.12	0.19	0.07	0.54	0.26	0.28	0.12

Table 4.6: Per-label performance of the **Qwen2.5-VL-3B-Instruct** model using only the *Answer* as input. Labels are evaluated on Precision (P), Recall (R), and F_1 -score under both global and local thresholding strategies. For each metric, the mean (M) and standard deviation (SD) over five cross-validation folds are reported.

Table 4.7 presents the performance of the same four inference configurations evaluated earlier, this time using the **all-MiniLM-L6-v2** sentence transformer for inference in place

of **Qwen2.5-VL-3B-Instruct**. All metrics are reported as mean $\pm$ standard deviation across five cross-validation folds.

Compared to Qwen, MiniLM achieves lower scores on most micro-averaged metrics, indicating weaker performance on frequent labels. However, it exhibits a modest but consistent gain in macro-averaged metrics—particularly when using local thresholds. This suggests that MiniLM provides more balanced per-class predictions, especially for less frequent labels.

Notably, when using local thresholds, MiniLM achieves a higher macro- F_1 than Qwen: 0.39 vs. 0.37 for the *Answer* input, and 0.38 vs. 0.33 for *Question [SEP] Answer*. These results support the hypothesis that MiniLM, as a sentence transformer specifically trained for capturing textual semantics, produces embeddings better suited for class-discriminative inference from generated text.

Additionally, in contrast to Qwen, MiniLM benefits from including the question text: the *Question [SEP] Answer* composition outperforms the *Answer*-only variant on several key metrics—both micro and macro—including F_1 and Jaccard similarity. When directly compared to Qwen’s *Question [SEP] Answer* variant, MiniLM improves on all metrics except micro-recall.

Input composition Threshold type	<i>Answer</i>				<i>Question [SEP] Answer</i>			
	Global		Local		Global		Local	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Precision (micro)	0.28	0.09	0.27	0.03	0.35	0.01	0.29	0.04
Precision (macro)	0.24	0.03	0.33	0.03	0.26	0.03	0.35	0.05
Recall (micro)	0.65	0.28	0.75	0.03	0.50	0.02	0.67	0.05
Recall (macro)	0.61	0.31	0.68	0.05	0.36	0.03	0.60	0.07
F_1 (micro)	0.35	0.02	0.40	0.03	0.41	0.01	0.41	0.04
F_1 (macro)	0.29	0.08	0.39	0.03	0.26	0.02	0.38	0.04
Jaccard (micro)	0.21	0.01	0.25	0.02	0.26	0.01	0.25	0.03
Jaccard (macro)	0.20	0.06	0.27	0.03	0.18	0.02	0.27	0.03
Hamming loss	0.44	0.16	0.41	0.02	0.26	0.04	0.36	0.02
Subset accuracy	0.01	0.01	0.01	0.01	0.02	0.02	0.00	0.01

Table 4.7: Performance of the **Qwen2.5-VL-3B-Instruct** model with inference performed using **all-MiniLM-L6-v2** over the VLM output, evaluated across five cross-validation folds using the metrics defined in Table 4.3. Four configurations are compared, based on two input compositions (*Answer* vs. *Question [SEP] Answer*) and two thresholding strategies (Global vs. Local). For each metric, the mean and standard deviation (SD) are reported. Bold values indicate the best-performing configuration per metric.

Table 4.8 presents per-label performance (mean $\pm$ SD over five folds) for the MiniLM

sentence-transformer using only the VLM’s generated answer as input. We compare global versus local thresholding.

We see that *Bottle Feeding* benefits from the use of MiniLM, with F_1 improving from 0.51 to 0.67 for local thresholds. However, the high standard deviation in both configurations indicates considerable variability across folds, suggesting that performance of this label is unstable. A plausible explanation is that each fold evaluates the model on samples drawn from different videos, and the model may overfit to the inter-video cosine similarity scores.

Metric	Global						Local					
	P		R		F_1		P		R		F_1	
	M	SD	M	SD	M	SD	M	SD	M	SD	M	SD
Flexing	0.06	0.09	0.33	0.47	0.10	0.15	0.04	0.10	0.15	0.34	0.07	0.15
Finger splay	0.26	0.41	0.57	0.49	0.15	0.08	0.13	0.04	0.89	0.11	0.23	0.05
Eye squeeze	0.11	0.07	0.37	0.31	0.17	0.12	0.14	0.03	0.71	0.25	0.24	0.05
Hand-to-face	0.53	0.16	0.90	0.13	0.64	0.09	0.73	0.06	0.79	0.06	0.75	0.04
Hand-to-mouth	0.38	0.14	0.95	0.07	0.53	0.13	0.48	0.05	0.83	0.05	0.61	0.05
Mouth open	0.33	0.30	0.40	0.37	0.36	0.33	0.67	0.13	0.63	0.18	0.63	0.09
Adult hand	0.12	0.11	0.48	0.45	0.19	0.17	0.24	0.04	0.99	0.02	0.39	0.05
Yawn	0.21	0.25	0.67	0.36	0.24	0.15	0.28	0.08	0.73	0.20	0.40	0.09
Bottle feeding	0.53	0.49	0.57	0.52	0.55	0.50	0.80	0.45	0.60	0.39	0.67	0.40
Grasping	0.08	0.05	0.62	0.52	0.13	0.09	0.08	0.04	0.74	0.36	0.15	0.07
Hand sucking	0.13	0.04	0.81	0.14	0.22	0.07	0.12	0.05	0.77	0.22	0.20	0.08
Tongue out	0.18	0.10	0.61	0.24	0.24	0.07	0.27	0.07	0.40	0.19	0.31	0.10

Table 4.8: Per-label performance of the Qwen2.5-VL-3B-Instruct model with inference performed using all-MiniLM-L6-v2, using only the *Answer* as input. Labels are evaluated on Precision (P), Recall (R), and F_1 -score under both global and local thresholding strategies. For each metric, the mean (M) and standard deviation (SD) over five cross-validation folds are reported.

4.3 Prompt Ensembling

The previous experiments demonstrated that VLMs can be effectively applied to multi-label classification tasks, achieving performance that exceeds baseline classifiers. However, we also observed substantial variability in performance across individual labels. Some cues (e.g. *Hand-to-Face*) were classified reliably, whereas others (e.g. *Flexing* and *Bottle Feeding*) exhibited either consistently low performance or high variance across folds. We hypothesize that a key bottleneck lies in the design and reliability of textual prompts. In particular, we identify four ways in which the current prompting pipeline may fail:

1. **Representation gap:** The VLM may not have encountered sufficient examples of a particular label or concept during pretraining, resulting in poor or inaccurate generated answers.

2. **Prompt specificity:** A prompt may be ill-defined or ambiguous, leading to weak alignment between the question and the desired semantic content.
3. **Cosine similarity overlap:** The cosine similarities between model-generated answers and reference templates may not clearly separate positive and negative cases, complicating threshold-based classification.
4. **Inconsistent generation:** The model does not fully generalize across input variations. The VLM may produce systematically different answer styles across video groups (e.g., due to visual or contextual bias), affecting the value and distribution of cosine similarities. This may be explained by high label-specific standard deviations.

While the first failure mode is intrinsic to the model’s training data and cannot be addressed without fine-tuning, the remaining three can potentially be mitigated through prompt ensembling.

Prompt ensembling involves generating multiple semantically equivalent prompts and aggregating their outputs. This method aims to:

- reduce the influence of any single poorly worded or ineffective prompt (mitigating point 2),
- smooth out the variance in cosine-similarity scores within positive and negative cases (mitigating point 3),
- increase model generalization by averaging across diverse outputs, decreasing variance between folds (mitigating point 4).

4.3.1 Methodology

For each label, we construct a set of five diverse prompts that all query the same underlying cue. Corresponding to the expected outputs of the model, we generate five template answers per label, phrased differently but semantically equivalent. Prompt and answer template pairs are presented in Table C.2, Appendix C.

During inference, the VLM is queried with all five prompts for a given input frame, yielding five textual outputs. Each of these outputs is embedded using a sentence transformer to produce a vector representation. Similarly, we compute embeddings for each of the five template answers. The five output embeddings are concatenated to form a single aggregated representation of the model’s response across prompts. The same concatenation is applied to the template answer embeddings. We then compute the cosine similarity between the aggregated output and template vectors to produce the final label-specific similarity score.

With this approach, we intend to increase the robustness of the final cosine-similarity scores and thresholds by averaging over multiple textual formulations. We hypothesize that this can help reduce the impact of prompt-specific or sample-specific biases. We expect that this will improve both overall performance and stability across cross-validation folds, particularly for labels that previously suffered from high variance. While incorporating multiple prompts introduces a broader range of variation, which could potentially introduce additional noise, we expect this risk to be mitigated by averaging over multiple embeddings as compared to using a single embedding. In doing so, we aim to capture a more stable and representative signal across diverse phrasings.

Prompt ensembling will be implemented on top of the MiniLM-based configuration, which uses only the VLM’s answer output for inference. To evaluate the effectiveness of the ensembling strategy, we compare the ensembled model’s performance against the original model using the same set of evaluation metrics, as outlined in Table 4.3.

4.3.2 Results

Tables 4.9 and 4.10 summarize the performance of the prompt ensembling approach using Qwen with answer-only input, evaluated under both global and local thresholding strategies.

Unlike in previous experiments—where global thresholding consistently yielded stronger micro-averaged metrics and local thresholding improved macro-averaged metrics, global thresholds now achieve the highest recall scores (both micro and macro), while local thresholds outperform in precision and F_1 metrics. Local thresholding also achieves better Jaccard scores, lower Hamming loss, and Subset Accuracy.

When compared to our previously best-performing model (MiniLM with answer-only input), the ensemble model improves only on micro-averaged precision. For all other metrics—including F_1 , using MiniLM and a single textual prompt remains superior, indicating that prompt ensembling does not yield general performance improvements in this configuration.

	Global		Local	
	Mean	SD	Mean	SD
Precision (micro)	0.23	0.04	0.29	0.06
Precision (macro)	0.24	0.03	0.30	0.03
Recall (micro)	0.79	0.07	0.69	0.06
Recall (macro)	0.74	0.08	0.57	0.09
F_1 (micro)	0.36	0.04	0.40	0.05
F_1 (macro)	0.33	0.02	0.34	0.02
Jaccard (micro)	0.22	0.03	0.25	0.04
Jaccard (macro)	0.22	0.02	0.24	0.02
Hamming Loss	0.53	0.04	0.38	0.07
Subset Accuracy	0.00	0.01	0.01	0.01

Table 4.9: Performance of the prompt-ensembled Qwen2.5-VL-3B-Instruct model with inference performed using all-MiniLM-L6-v2 using only the *Answer* as input, evaluated across five cross-validation folds using the metrics defined in Table 4.3. Global and local thresholding strategies are compared. For each metric, the mean and standard deviation (SD) are reported. Bold values indicate the best-performing configuration per metric.

Table 4.10 presents per-label metrics across global and local thresholds. Among all labels, only *Finger Splay* shows notable improvement in F_1 score, increasing from 0.23 to 0.28 under local thresholding when compared to single question answer only MiniLM. This suggests that the original prompt-template pair for this label may have poorly captured the semantic distinction between positive and negative instances of the label, and that ensembling mitigated this issue by averaging over multiple answers.

For the remaining labels, performance remained unchanged or declined. A possible explanation is that while ensembling is expected to reduce variance, it may also blur the distinction between present and absent cases. Instead of forming two distinct clusters in cosine similarity space, embeddings of positive and negative samples may converge into a single, less informative group. In effect, prompt ensembling may have produced the opposite of the intended outcome: reducing, rather than enhancing, semantic contrast.

Furthermore, the standard deviation for *Bottle Feeding* remains high under both global and local thresholding, despite the use of ensembling. This indicates that the variance across folds is unlikely due to inconsistencies in phrasing across video groups. Instead, it suggests that the VLM systematically struggles to classify samples from a particular video.

Metric	Global						Local					
	P		R		F_1		P		R		F_1	
	M	SD	M	SD	M	SD	M	SD	M	SD	M	SD
Flexing	0.04	0.06	0.17	0.24	0.06	0.09	0.01	0.02	0.20	0.45	0.02	0.04
Finger splay	0.12	0.04	1.00	0.00	0.22	0.06	0.32	0.29	0.27	0.20	0.28	0.23
Eye squeeze	0.12	0.05	0.53	0.29	0.19	0.09	0.11	0.07	0.50	0.47	0.17	0.11
Hand-to-face	0.53	0.10	0.92	0.06	0.67	0.08	0.57	0.09	0.83	0.07	0.67	0.06
Hand-to-mouth	0.44	0.06	0.54	0.17	0.48	0.09	0.44	0.11	0.75	0.13	0.55	0.12
Mouth open	0.45	0.07	0.95	0.03	0.61	0.06	0.58	0.11	0.74	0.12	0.64	0.07
Adult hand	0.21	0.03	0.72	0.14	0.32	0.03	0.24	0.04	0.99	0.03	0.39	0.05
Yawn	0.11	0.02	1.00	0.00	0.20	0.04	0.12	0.08	0.35	0.23	0.18	0.12
Bottle feeding	0.47	0.32	0.67	0.40	0.53	0.31	0.72	0.42	0.53	0.36	0.59	0.35
Grasping	0.10	0.03	0.96	0.09	0.18	0.05	0.10	0.05	0.75	0.28	0.17	0.08
Hand sucking	0.15	0.09	0.55	0.28	0.23	0.13	0.10	0.05	0.57	0.30	0.17	0.08
Tongue out	0.14	0.02	0.92	0.13	0.24	0.03	0.28	0.13	0.38	0.23	0.30	0.16

Table 4.10: Per-label performance of the prompt-ensembled **Qwen2.5-VL-3B-Instruct** model with inference performed using **all-MiniLM-L6-v2**, using only the *Answer* as input. Labels are evaluated on Precision (P), Recall (R), and F_1 -score under both global and local thresholding strategies. For each metric, the mean (M) and standard deviation (SD) over five cross-validation folds are reported.

5 Discussion

This study explored the effectiveness of Vision-Language Models (VLMs) in classifying behavioral cues in video recordings of preterm infants. Specifically, we investigated two research questions: (1) *How effective are Vision-Language Models at classifying behavioral cues in video recordings of preterm infants?* and (2) *What impact do different compositional and linguistic design choices of input prompts and answer templates have on the classification performance of the Vision-Language Model?*. In the following section, we interpret our findings in light of these questions and outline key limitations and directions for future research.

5.1 Overall Performance and Key Findings

To address the first research question: *How effective are Vision-Language Models at classifying behavioral cues in video recordings of preterm infants?* — we evaluated the performance of **Qwen2.5-VL-Instruct** using a zero-shot, text-based classification pipeline. The question is motivated by the need of non-invasive monitoring techniques in the NICU, with the ability to assess the infants state (e.g. hunger, pain, discomfort, sleep) as the endmost goal. Our experiments demonstrate that VLMs can identify certain behavioral cues with moderate success. The best performing configuration, which uses **Qwen2.5-VL-3B-Instruct** for generation and **all-MiniLM-L6-v2** for language-based inference on the VLM’s output, achieved a macro- F_1 score of 0.39 ± 0.03 . This pipeline outperforms all baseline classifiers, with the random classifier yielding the highest among them (macro- $F_1 = 0.25 \pm 0.02$). This confirms that the VLM is capable of picking up on meaningful visual-semantic cues, indicating that classification using VLMs is feasible in this context.

However, performance was uneven across labels. Certain cues, such as *Hand-to-Face*, *Hand-to-Mouth*, and *Mouth Open* were recognized frequently. In contrast, other cues such as *Flexing*, *Finger Splay*, *Eye Squeeze*, *Adult Hand*, *Yawn*, *Grasping*, *Hand Sucking*, and *Tongue Out* were poorly classified. This discrepancy likely arises from a combination of factors. Many of the poorly performing cues, such as *Hand Sucking* and *Finger Splay*, are specific to infants and likely underrepresented or entirely absent in the VLM’s pretraining data. In contrast, cues with close analogs in adult behavior (e.g., *Mouth Open*) may generalize more readily. Furthermore, some behaviors, such as *Flexing*, may manifest differently in infants compared to adults, making domain transfer more difficult. Finally, several cues like *Grasping* and *Yawn* unfold over time and may be harder to interpret from a single static frame, further complicating classification in a frame-based pipeline.

Interestingly, our results from the embedding visualization experiment and the text-based

classification pipeline revealed that visual-semantic clustering in the embedding space does not always entail high classification performance. For instance, *Bottle Feeding* and *Adult Hand* showed distinct clustering but yielded low F_1 -scores. A plausible explanation is that visual similarity drives the clustering, rather than true semantic understanding. The VLM may group similar-looking images without truly “understanding” the behavioral cue in question. Another contributing factor could be the limitations of the text-based classification pipeline itself. Performance is not solely determined by the model’s internal representations, but also depends on the phrasing of both the input prompt and the answer template. Furthermore, the language inference component must effectively distinguish between positive and negative classes in embedding space, an aspect that may vary in robustness across different labels.

The high variance observed for *Bottle Feeding* suggests that this class was recognized successfully in some test folds but poorly in others. This inconsistency likely reflects variability in the source videos, such as differences in lighting, camera angle, or infant positioning. Expanding the dataset with more diverse infant videos would provide a clearer and more reliable estimate of this label’s true performance.

5.2 Influence of Prompt Design on Classification Performance

To address the second research question: *What impact do different compositional and linguistic design choices of input prompts and answer templates have on the classification performance of the VLM?* — we primarily draw on insights from the embedding visualization and prompt ensembling experiments.

The embedding visualization experiment demonstrated that prompt design influences the structure of the VLM’s embedding space. In particular, closed-form prompts (e.g., yes/no questions) produced more distinct clusters for certain labels compared to open-ended prompts or no prompts at all. This suggests that prompting helps guide the model’s attention toward relevant visual features. Although prompts did introduce some attraction between different infant videos, the most prominent clustering occurred within individual videos. This implies that although prompt design affects the model’s representations, these are still predominantly shaped by visual information propagated through the vision encoder.

In the prompt ensembling experiment, we averaged predictions across multiple prompt formulations. While this approach did not lead to improvements in overall model performance, it did improve label classification performance for *Finger Splay*, of which the F_1 -score increased by 0.06 with ensembling. This indicates that the original single-prompt

formulation for this cue may have been suboptimal. Likely because the term ‘splaying’ is uncommon or absent in training data. For the other cues, it is plausible that ensembling reduced the separability of cosine-similarities between positive and negative samples, either by diluting the cosine-similarities across the different prompts or by introducing more noise through poorly formulated ones, rather than reducing it.

Taken together, these findings suggest that both composition of the prompt and formulation can be bottlenecks in leveraging VLMs for classification. While careful prompt engineering can improve model performance, manually optimizing prompts is time-consuming, difficult to scale, and prone to diminishing returns, particularly for subtle cues.

5.3 Limitations

The primary limitations of this study stem from the characteristics of the SLAPI dataset. Only five annotated videos were available, and even after excluding critically underrepresented labels, the distribution remains highly imbalanced. This data sparsity contributes to high standard deviations in performance metrics for rare labels. This makes it difficult to confidently assess the performance of these labels. Furthermore, it is difficult to verify whether this variability reflects true differences in model performance across folds or is simply a consequence of insufficient sample size. The limited amount of data makes the combination of strict video separation and balanced label distribution across folds impossible. Given the limited and uneven dataset, trade-offs had to be made between these competing goals.

Another notable limitation concerns the annotation quality. Annotations were performed by research interns at the UMC and subsequently reviewed by the author of this thesis. However, as of the time of writing, no standardized guidelines or formal training protocols were established for labeling the behavioral cues. This lack of standardized procedures raises concerns about the reliability and consistency of the ground truth annotations. Consequently, evaluation of the VLM may suffer from bias and ambiguous interpretations of cues.

In addition to annotation quality, the design of prompts and answer templates represents a further limitation. These were manually constructed by the author and, while efforts were made to maintain structural consistency across prompts and templates, the process of prompt ideation was inherently subjective. Although this study experimented with varying prompt structures and applied prompt ensembling techniques, it did not systematically explore the effects of specific keywords, linguistic phrasing, semantic nuance, and other factors that could influence model performance.

Finally, the classification pipeline relies on cosine similarity thresholding, treating the

model’s output as a binary prediction of label presence or absence. This approach does not account for ambiguous or uncertain responses, nor does it handle cases where the VLM output fails to address the prompt meaningfully. In such instances, the model’s response is still forced into a binary decision based on whichever template yields the highest similarity score, which may not reflect the true intent or confidence of the model.

5.4 Future Directions

Although the best-performing model in this study does not yet meet the performance that would be required for clinical implementation, perfect classification may not be necessary for clinical utility. Behavioral states are often characterized by the co-occurrence of multiple cues within a short time span. As such, partial overlap with the ground truth may be sufficient in practical settings. In order to reach a more suitable performance, model robustness could be further improved by leveraging temporal information through the sampling of consecutive video frames. Since adjacent frames often share similar label sets and infant states, comparing temporally adjacent outputs could help reduce the impact of outliers and enhance reliability. Additionally, since behavioral cues frequently co-occur, as shown in Figures 3.1 and 3.2, recognizing specific cue combinations may offer an alternative route for improving classification accuracy. Integrating additional modalities that were discussed in Section 2.2, may also increase robustness.

Beyond the research direction of this thesis, the availability of more extensive NICU data would open several possibilities for VLM training and fine-tuning. Because prompt engineering plays a critical role in the performance of the current pipeline, future work could explore soft prompting techniques as discussed in Section 2.4.2. These allow for trainable prompts without requiring full model fine-tuning [Zhou et al., 2022c], offering a more efficient alternative. However, it remains to be seen whether this would yield substantial improvements or whether performance is primarily constrained by the VLM’s inherent domain knowledge. In this case, VLM performance could still benefit from regular fine-tuning.

Looking ahead, even as VLMs continue to improve with larger and more diverse training data, NICU data may lag behind due to privacy constraints and limited availability. This likely results in a decrease in performance for underrepresented data domains. Future models may bridge this by exploring domain adaptation strategies, aiming to transfer knowledge from a source domain (e.g. adult imagery or infants outside of healthcare-related settings) to a target domain (e.g. the NICU). Since infant behavioral cues don’t necessarily parallel adult cues, effective domain translation will require better intermediary representations. Methods for domain adaptation [Zhou et al., 2022a] and the approach by Saha et al. [2024], that enriches fine-grained label descriptions, could help reveal implicit

similarities between cues in the source and target domains.

Lastly, many behavioral cues are temporally defined and not easily identifiable in static frames. These issues can be resolved with the development of video-based VLMs. However, although video-based classification could address this, video based classification demands more training data and is computationally more expensive. Consequently, such approaches may only become feasible as inference costs decrease, and previously mentioned challenges regarding NICU data-scarcity are mitigated.

6 Conclusion

This thesis explored the feasibility of leveraging the large, pretrained latent space of Vision-Language Models (VLMs) to classify behavioral cues in preterm infants based on video recordings. The motivation behind this work lies in the potential to enable non-invasive monitoring of infant states — such as sleep, hunger, and pain — which could ultimately improve clinical outcomes for this vulnerable population.

Our findings demonstrate that, even without domain-specific training, VLMs exhibit promising capabilities in zero-shot classification of visually salient cues, particularly when guided by well-structured textual prompts. By leveraging the multimodal embedding space of Qwen2.5-VL, we showed that language can enhance cue recognition by steering visual attention and structuring embeddings more semantically.

Although the current system does not yet achieve the level of precision required for autonomous infant monitoring, its ability to encode relevant semantic distinctions suggest potential for supporting neonatal behavioral assessment. Importantly, the prompt-based framework provides a degree of explainability uncommon in conventional vision-only models, allowing clinicians and researchers to understand and trace how specific cues are detected or overlooked.

Several limitations, including the reliance on manual prompt ideation and the absence of fine-tuning, constrain the generalizability of the current findings. However, these also highlight critical avenues for future work. These include developing more robust and learnable prompting strategies, incorporating additional modalities, and fine-tuning VLMs on NICU-specific data.

In conclusion, this research provides a foundation for applying VLMs in neonatal care and presents insights for further development. By bridging visual observations with linguistic reasoning, VLMs may contribute to more scalable, interpretable, and non-invasive monitoring solutions in the NICU.

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A Cross-Validation Pseudocode

Algorithm 1: SoftGroupedIterativeStratification($D, n, r_1, \dots, r_n, \epsilon$), adapted from Sechidis et al. [2011]

Input: A set of instances D , annotated with labels $L = \{\lambda_1, \dots, \lambda_q\}$ and group number $g \in G = \{1, \dots, h\}$, desired amount of grouping slack ϵ , desired number of subsets k , desired proportion of examples in each subset, $r_1, \dots, r_k$ (e.g. in 10-fold CV with 5 groups $k = 10, r_j = 0.1, j = 1 \dots 10, h = 1 \dots 5$)

Output: Disjoint subsets $S_1, \dots, S_k$ of D

```

// Calculate the desired number of examples at each subset
1 for  $j \leftarrow 1$  to  $k$  do
2    $c_j \leftarrow |D|r_j$ 
   // Initialise group count at each subset to 0
3   for  $h \leftarrow 1$  to  $|G|$  do
4      $n_j^h \leftarrow 0$ 
   // Calculate the desired number of examples of each label at each subset
5 for  $i \leftarrow 1$  to  $|L|$  do
   // Find the examples of each label in the initial set
6    $D^i \leftarrow \{(\mathbf{x}, Y, g) \in D : \lambda_i \in Y\}$  for  $j \leftarrow 1$  to  $k$  do
7      $c_j^i \leftarrow |D^i|r_j$ 
8 while  $|D| > 0$  do
   // Find the label with the fewest (but at least one) remaining examples,
   // breaking ties randomly
9    $D^i \leftarrow \{(\mathbf{x}, Y, g) \in D : \lambda_i \in Y\}$ 
10   $l \leftarrow \operatorname{argmin}_i (|D^i|) \cap \{i : D^i \neq \emptyset\}$  foreach  $(\mathbf{x}, Y, g) \in D^l$  do
   // Find the subset(s) with a label deficiency exceeding the largest
   // deficiency of this label minus slack.
11   $M \leftarrow \{j = 1 \dots k : c_j^l \geq \max_{i=1 \dots k} c_i^l - \epsilon\}$ 
   // breaking ties by considering the number of instances of the same group,
   // breaking further ties by considering subset capacity, breaking further
   // ties randomly.
12  if  $|M| = 1$  then
13     $m \in M$ 
14  else
15     $M' \leftarrow \operatorname{argmax}_{j \in M} (n_j^g)$ 
16    if  $|M'| = 1$  then
17       $m \in M'$ 
18    else
19       $M'' \leftarrow \operatorname{argmax}_{j \in M} (c_j)$ 
20      if  $|M''| = 1$  then
21         $m \in M''$ 
22      else
23         $m \leftarrow \operatorname{randomElementOf}(M'')$ 
24   $S_m \leftarrow S_m \cup \{(\mathbf{x}, Y, g)\}$ 
25   $D \leftarrow D \setminus \{(\mathbf{x}, Y, g)\}$ 
   // Update group subsets, instances, and group counts
26   $n_m^g \leftarrow n_m^g + 1$ 
   // Update desired number of examples
27  foreach  $\lambda_i \in Y$  do
28     $c_m^i \leftarrow c_m^i - 1$ 
29   $c_m \leftarrow c_m - 1$ 
30 return  $S_1, \dots, S_k$ 

```

B Cue Labeled t-SNE's Embeddings

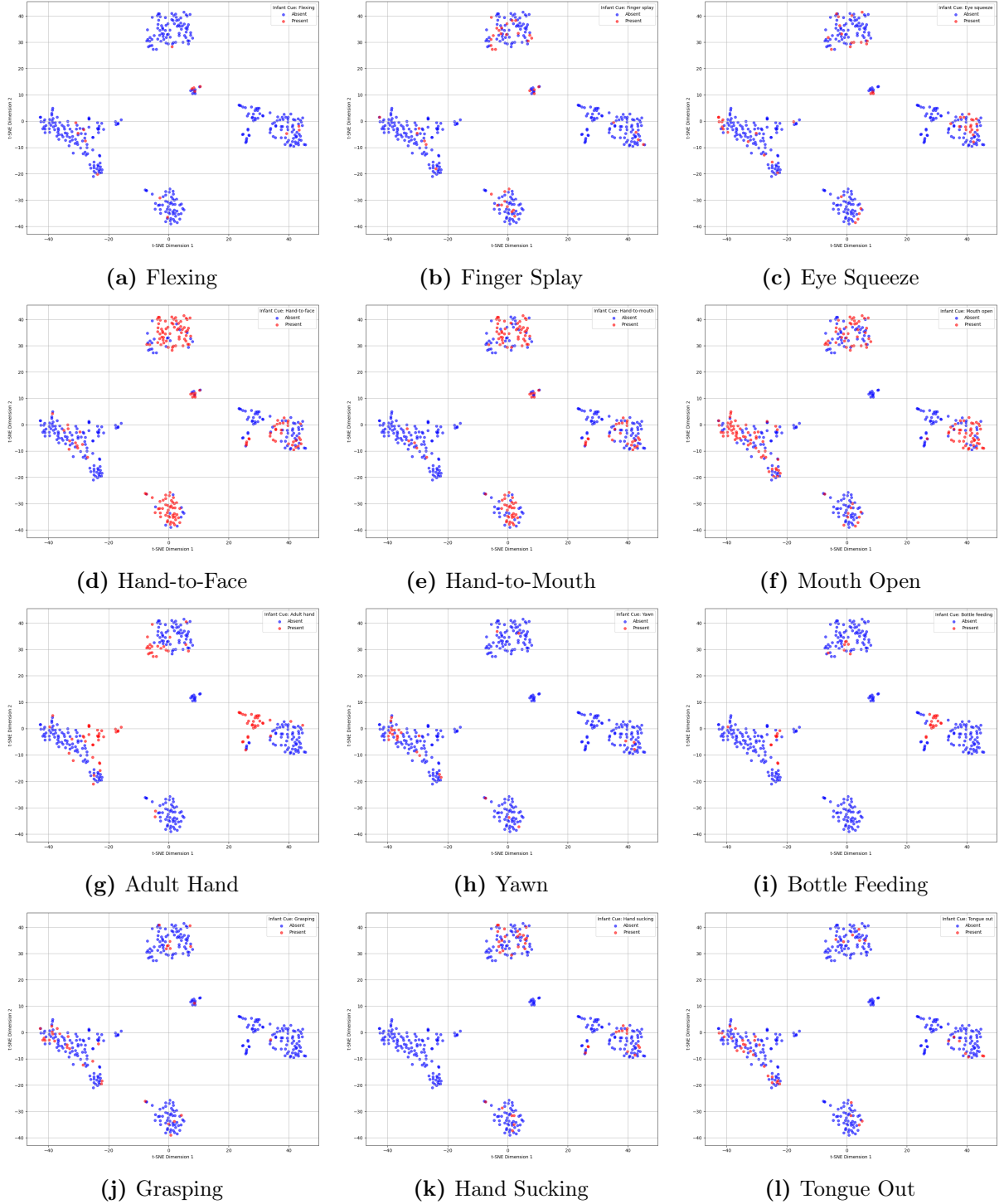


Figure B.1: t-SNE embeddings of annotated samples across all infants. The embeddings are obtained by forwarding each sample frame without a prompt through the VLM. The samples are colored based on whether the cue is present (Red) or not (Blue).

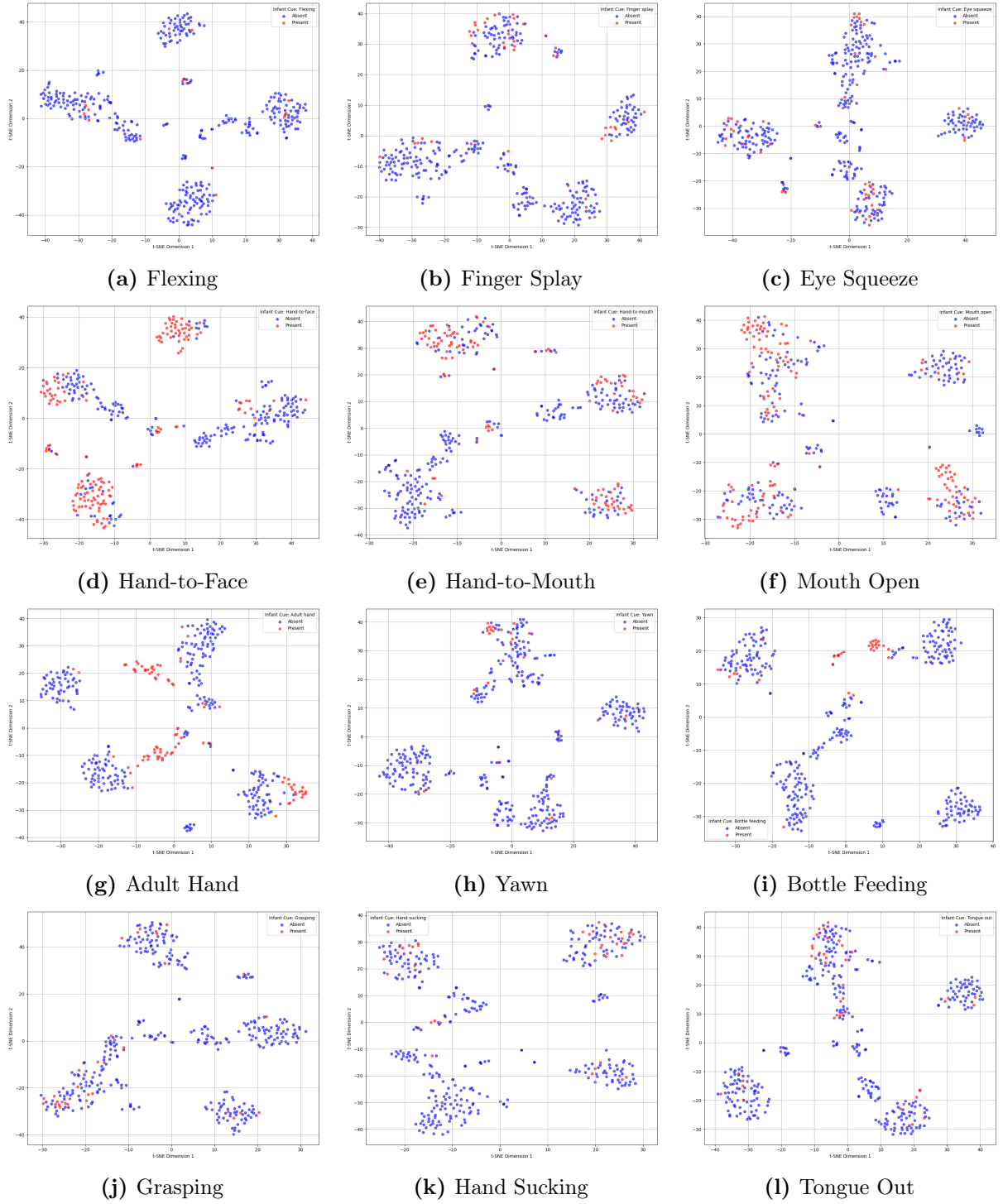


Figure B.2: t-SNE embeddings of annotated samples across all infants. The embeddings are obtained by forwarding each sample frame and their corresponding closed question (see Table 4.2) through the VLM. The samples are colored based on whether the cue is present (Red) or not (Blue).

C Prompts and Answer Templates

#	Label	Question	Answer Template
1	Flexing	<i>Is the infant flexing its body?</i>	<i>Yes, the infant appears to be flexing its body</i>
2	Finger Splay	<i>Is the infant splaying its fingers?</i>	<i>Yes, the infant appears to be splaying its fingers</i>
3	Eye Squeeze	<i>Is the infant squeezing its eyes?</i>	<i>Yes, the infant appears to be squeezing its eyes</i>
4	Hand-to-Face	<i>Is the infant's hand touching its face?</i>	<i>Yes, the infant appears to be touching its face with its hand</i>
5	Hand-to-Mouth	<i>Is the infant's hand touching its mouth?</i>	<i>Yes, the infant appears to be touching its mouth with its hand</i>
6	Mouth Open	<i>Is the infant's mouth open wide?</i>	<i>Yes, the infant's mouth appears to be open</i>
7	Adult Hand	<i>Is there an adult hand visible in the image?</i>	<i>Yes, there is an adult hand visible in the image</i>
8	Yawn	<i>Is the infant yawning?</i>	<i>Yes, the infant appears to be yawning</i>
9	Bottle Feeding	<i>Is the infant being bottle fed?</i>	<i>Yes, the infant appears to be bottle fed</i>
10	Grasping	<i>Is the infant grasping (onto something with) its hand?</i>	<i>Yes, the infant appears to be grasping onto something with its hand</i>
11	Hand Sucking	<i>Is the infant sucking on its fingers or the back of its hand?</i>	<i>Yes, the infant appears to be sucking its hand</i>
12	Tongue Out	<i>Is the infant sticking its tongue out?</i>	<i>Yes, the infant appears to be sticking its tongue out</i>

Table C.1: List of the closed question prompts submitted to the model alongside the frame of the annotated sample. The model output is compared with the subsequent answer template. Each question aims to assess the presence of a single class.

#	Label	Question	Answer Template
1	Flexing	<i>Is the infant flexing its body?</i> <i>Do you see the baby tensing its limbs?</i> <i>Does the child look like it's in a flexed position?</i> <i>Does the baby appear to be tensing up?</i> <i>Do you notice the infant actively flexing its muscles</i>	<i>Yes, the infant appears to be flexing its body</i> <i>Yes, the baby seems to be tensing its limbs</i> <i>Yes, the child is flexing its muscles</i> <i>Yes, the baby is visibly tensing up</i> <i>Yes, the infant's limbs are flexed</i>
2	Finger Splay	<i>Is the infant splaying its fingers?</i> <i>Do the baby's fingers look spread apart?</i> <i>Can you see the child's fingers spread out?</i> <i>Does it look like the baby's fingers are splayed?</i> <i>Are the child's fingers extended?</i>	<i>Yes, the infant appears to be splaying its fingers</i> <i>Yes, the baby's fingers are spread apart</i> <i>Yes, you can see the child's fingers being spread out</i> <i>Yes, the infant's fingers are splayed</i> <i>Yes, the child's fingers are extended</i>
3	Eye Squeeze	<i>Is the infant squeezing its eyes?</i> <i>Do you notice the baby squinting its eyes?</i> <i>Does it look like the child is tightly closing its eyes?</i> <i>Are the infant's eyelids firmly squeezed together?</i> <i>Can you see the baby squeezing its eyes?</i>	<i>Yes, the infant appears to be squeezing its eyes</i> <i>Yes, one can see the baby squinting their eyes</i> <i>Yes, the child's eyes are tightly closed</i> <i>Yes, the infant's eyelids are firmly squeezed together</i> <i>Yes, the baby is squeezing its eyes shut</i>
4	Hand-to-Face	<i>Is the infant's hand touching its face?</i> <i>Can you see the baby's hand on its face?</i> <i>Does the child have its hand planted against its cheek?</i> <i>Is the infant pressing its palm or fingers onto its face?</i> <i>Do you notice the baby resting its hand on its face?</i>	<i>Yes, the infant appears to be touching its face with its hand</i> <i>Yes, the baby has its hand placed on its face</i> <i>Yes, the baby's hand is resting on its cheek</i> <i>Yes, I can see the child pressing their palm against their face</i> <i>Yes the baby is resting its hand on its face</i>
5	Hand-to-Mouth	<i>Is the infant's hand touching its mouth?</i> <i>Do you see the baby placing its hand near its lips?</i> <i>Is the child's hand at or in its mouth?</i> <i>Can you spot the infant bringing its hand up to its mouth?</i> <i>Does it look like the baby is putting its hand against its lips?</i>	<i>Yes, the infant appears to be touching its mouth with its hand</i> <i>Yes, the baby's hand is at its lips</i> <i>Yes, the child seems to have its hand at or in its mouth</i> <i>Yes, the infant is bringing its hand to its mouth</i> <i>Yes, I can see the baby's hand against its lips</i>
6	Mouth Open	<i>Is the infant's mouth open wide?</i> <i>Do you see the baby with its mouth agape?</i> <i>Does the child have its mouth open?</i> <i>Is the infant's jaw dropped, opening its mouth?</i> <i>Can you notice the baby's lips being parted?</i>	<i>Yes, the infant's mouth appears to be open</i> <i>Yes, the baby's mouth is wide agape</i> <i>Yes, the child's mouth is open</i> <i>Yes, the infant looks like it is opening its mouth</i> <i>Yes, the baby's lips are parted and the mouth is open</i>
7	Adult Hand	<i>Is there an adult hand visible in the image?</i> <i>Do you see a grown-up's hand in the frame?</i> <i>Can you spot an adult hand near the infant?</i> <i>Is a caregiver's hand visible alongside the baby?</i> <i>Are there arms or hands of a nurse visible in the image?</i>	<i>Yes, there is an adult hand visible in the image</i> <i>Yes, I can see a grown-ups hand in the frame</i> <i>Yes, the adult's hand is near the infant</i> <i>Yes, the adult's hand is in view alongside the baby</i> <i>Yes, a nurse or adult hand is visible in the image</i>
8	Yawn	<i>Is the infant yawning?</i> <i>Can you see the baby opening its mouth in a yawn?</i> <i>Does it look like the child is yawning?</i> <i>Is the infant's mouth wide open, yawning?</i> <i>Do you notice the baby stretching the jaw as if yawning?</i>	<i>Yes, the infant appears to be yawning</i> <i>Yes, the baby's mouth is open in a yawn</i> <i>Yes, I can see the infant yawning</i> <i>Yes, the baby looks like it's mid-yawn</i> <i>Yes, the child seems to be stretching its jaw in a yawn</i>
9	Bottle Feeding	<i>Is the infant being bottle fed?</i> <i>Is there a bottle visible in the image?</i> <i>Is the child drinking from a bottle?</i> <i>Can you spot a bottle in the infant's mouth?</i> <i>Does the image show the baby with a bottle?</i>	<i>Yes, the infant appears to be bottle fed</i> <i>Yes there is a bottle visible in the image</i> <i>Yes, the child is drinking from a feeding bottle</i> <i>Yes, the infant's mouth is in contact with a bottle</i> <i>Yes, the image shows the baby feeding from a bottle</i>
10	Grasping	<i>Is the infant grasping (onto something with) its hand?</i> <i>Do you see the baby holding onto an object?</i> <i>Is the child's fist wrapped, possibly holding something?</i> <i>Is the infant's hand clenched into a fist?</i> <i>Is the infant's hand open or closed?</i>	<i>Yes, the infant appears to be grasping onto something with its hand</i> <i>Yes, the baby is holding an object</i> <i>Yes, I can see the child's fist wrapped around something</i> <i>Yes, the infant is clutching an item</i> <i>The infant's hand is closed</i>
11	Hand Sucking	<i>Is the infant sucking on its fingers or the back of its hand?</i> <i>Do you see the baby putting its hand in its mouth?</i> <i>Can you see the child's hand in its mouth, possibly sucking on it?</i> <i>Can you spot the baby with its hand partly inside its mouth?</i> <i>Does it appear the infant is sucking on its own hand?</i>	<i>Yes, the infant appears to be sucking its hand</i> <i>Yes, I can see the infant's hand in its mouth</i> <i>Yes, the baby's hand is in its mouth sucking</i> <i>Yes, it looks like the baby is having its hand in its mouth</i> <i>Yes, the child seems to be sucking on its own hand</i>
12	Tongue Out	<i>Is the infant sticking its tongue out?</i> <i>Do you see the baby's tongue extended?</i> <i>Does the child have its tongue protruding?</i> <i>Can you spot the infant's tongue sticking outward?</i> <i>Does it look like the baby is playfully sticking out its tongue?</i>	<i>Yes, the infant appears to be sticking its tongue out</i> <i>Yes, the baby's tongue is extended</i> <i>Yes, the child's tongue is protruding</i> <i>Yes, I can see the infant's tongue sticking outward</i> <i>Yes, it looks like the baby is playfully sticking out its tongue</i>

Table C.2: List of the closed question prompts submitted to the model alongside the frame of the annotated sample. The model output is compared with the subsequent answer template. The cosine similarities of each answer pair are ensembled. Each question aims to assess the presence of a single class.